Oleo-hydraulic components (II)

Universidad de Oviedo
Área de Mecánica de Fluidos

Escuela Politécnica de Ingeniería (EPI) de Gijón
4º curso – Grado en Ingeniería Mecánica

FLUID POWER: HYDRALIC AND PNEUMATIC TECHNOLOGIES

Chapter 2

https://www.innova.uniovi.es/innova/campusvirtual/campusvirtual.php

Outline.

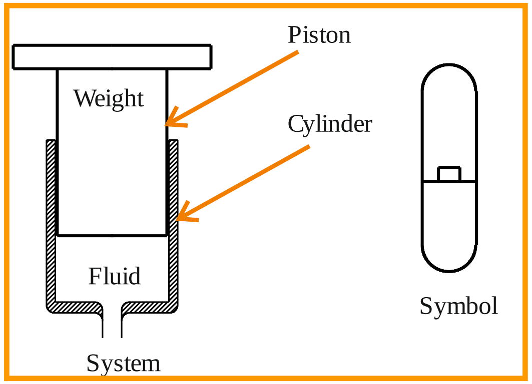
2.3. Accumulators.

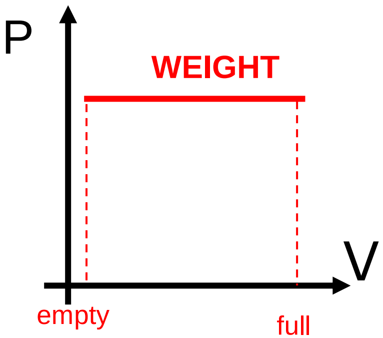
Definition and functions.

- This devices are used as **ENERGY ACCUMULATORS** by storing a fluid at a given **PRESSURE**.
- Their two basic functions are:
  - **AUXILIARY ENERGY SOURCE**:
    - They can deliver flow rate, to help the pump in feeding the system in the case of important flow rate variations along a given working cycle.
    - They might keep a given pressure in the system: presses.
  - **TRANSIENT DAMPER**:
    - Overpressures, water hammer, flow rate fluctuations in pumps, systems, ….

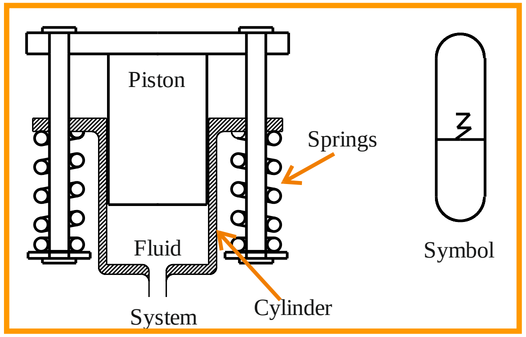
Types of accumulators (I).

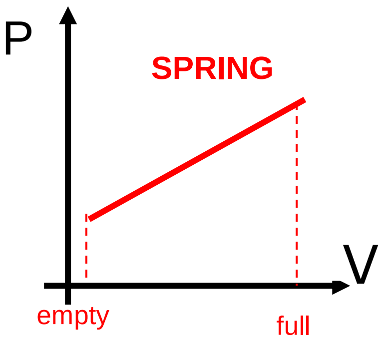
- **WEIGHT (Big volumes)**
  - The pressure is constant with time (filling evolution): $P=F/S=mg/S=constant$
  - They work due to the gravity force: always in vertical position and quite big elements.



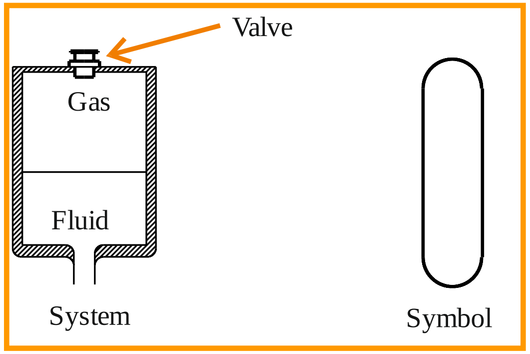


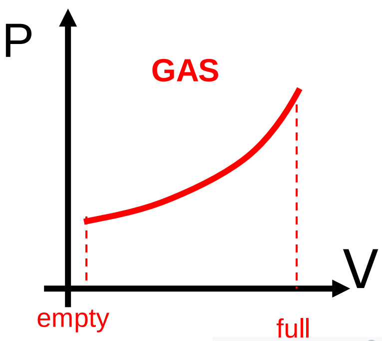
- **SPRING**
  - They work against a spring. The pressure changes with the movement: $P=F/S=kx/S=kV/S^2=f(\vartheta)$





- **GAS**
  - They work compressing a gas. They are more delicate as the gas might have some reaction with the oil passing through the leaks and some bubbles or cavitation might appear.

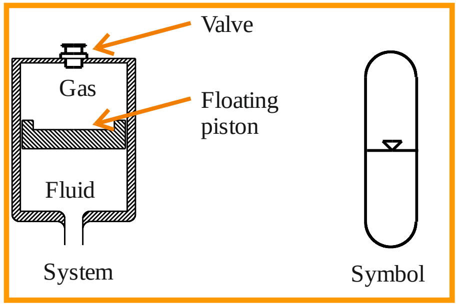


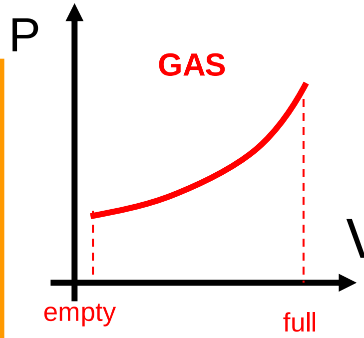


2.3. Accumulators.

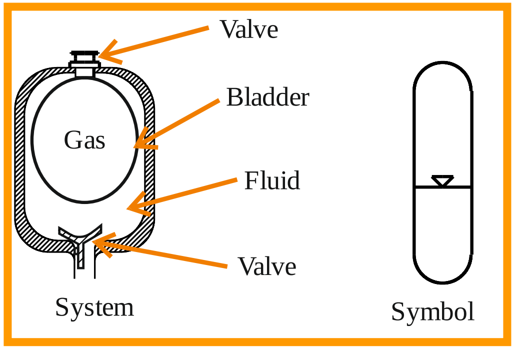
Types of accumulators (II).

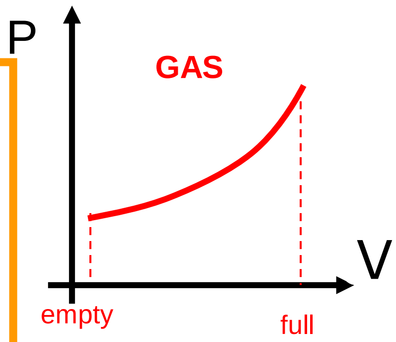
- ❑ **FLOATING PISTON**
  - ❑ To avoid the direct contact with the gas, as in the previous type. Even with this system, some leakages are possibly found, and problems with the oil may appear.



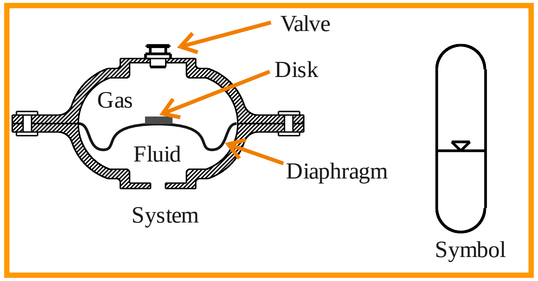


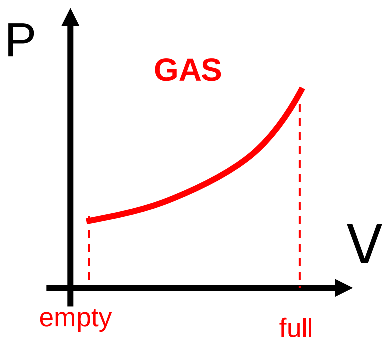
- ❑ **BLADDER (up to 50 litres)**
  - ❑ It works with big volumes.
  - ❑ The gas is isolated in the interior (bladder) made of neoprene. In this way, leakages are avoided.
  - ❑ PV^n = constant (n=1, isotherm; n=1.4, adiabatic)





- ❑ **DIAPHRAGM (small volumes)**
  - ❑ The direct contact is avoided by using a flexible diaphragm.



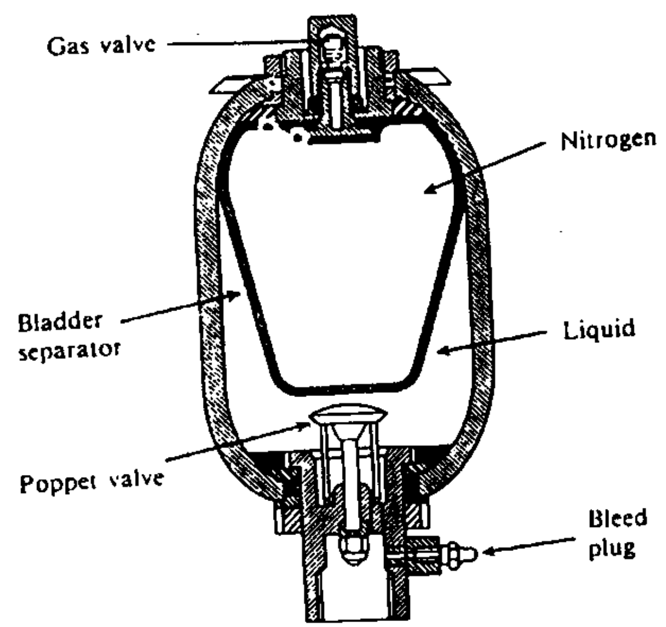


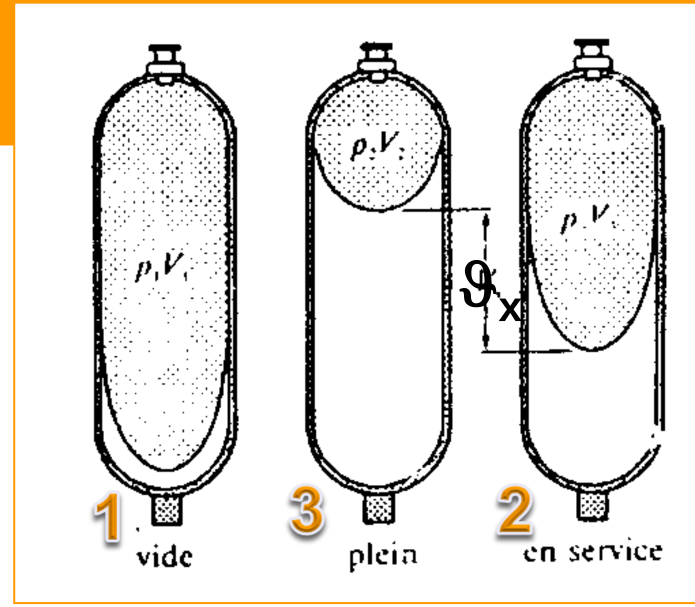
Types of accumulators (III).
Gas (nitrogen) accumulators.

BLADDER ACCUMULATOR (detail)

The pressure variation in the GAS accumulators is defined using the **Boyle-Mariotte law**: **PV^n = constant** (compression/ expansion of the gas), with n being:

- **n=1** → **ISOTHERMAL** evolution (very slow), for **t > 2 min** (filling).
- **n=1.4** → **ADIABATIC** evolution (very quick), for **t < 10 s** (emptying).
- **n=1.25** → **POLYTROPIC** evolution (intermediate between the previous).





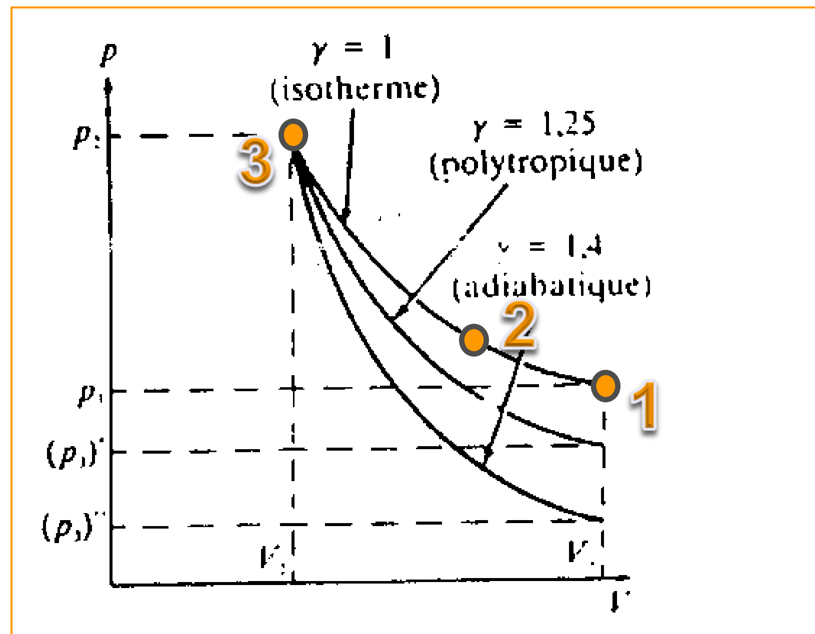
Types of accumulators (IV). Gas (nitrogen) accumulators. Size.

$$V_{ac} = V_x \frac{\left(\dfrac{p_{min}}{p_{ac}}\right)^{1/n}}{\left[1 - \left(\dfrac{p_{min}}{p_{max}}\right)^{1/n}\right]}$$

MIND: Absolute pressure !! (Thermodynamic calculation, sum 1 bar)

$$\frac{p_{min}}{p_{ac}} = \frac{1}{0.9}$$

- **$V_{ac} = V_1$: Accumulator volume** (gas volume at the initial pressure).
- **$P_{ac} = p_1$: Initial filling pressure** (usually, a 90% of the minimum emptiness pressure).
- **$P_{max} = p_3$: Maximum working pressure.**
- **$P_{min} = p_2$: Minimum working pressure.**
- **V_x: Oil volume (incompressible)**



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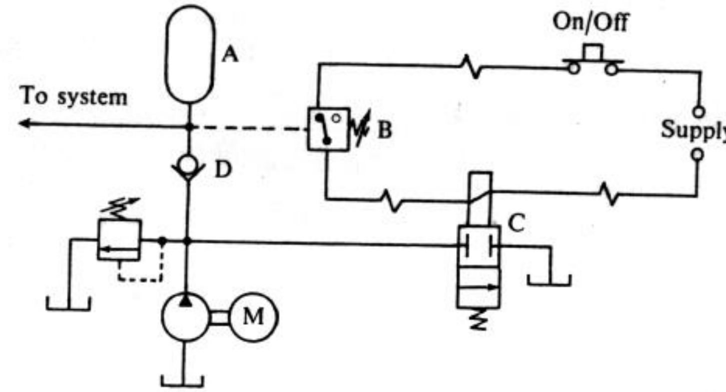
2.3. Accumulators.



Exercise 4:

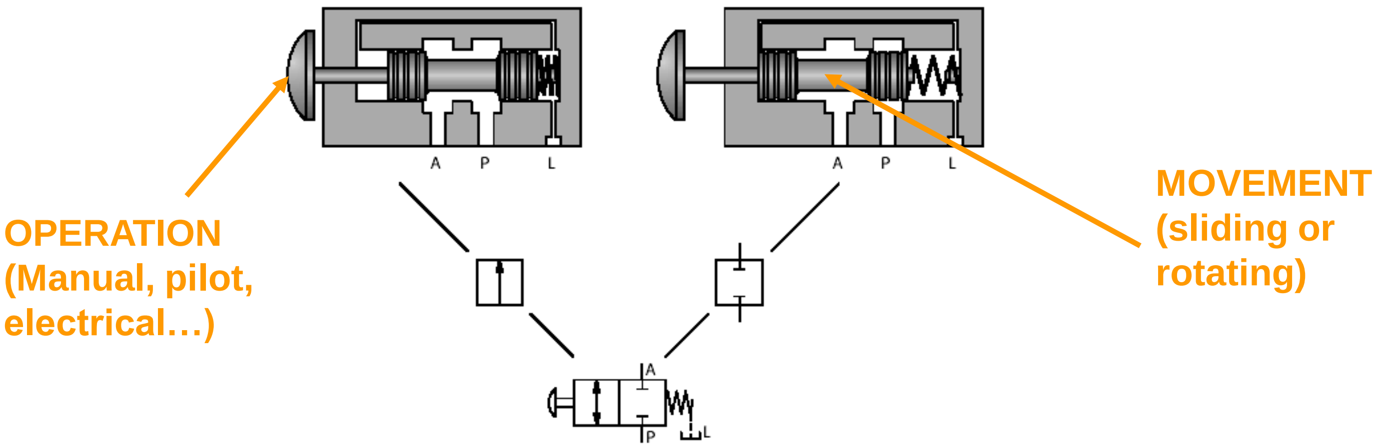
UO: UOxxxHTU

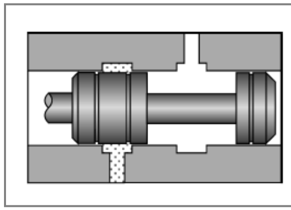
An oleo-hydraulic pump delivers flow rate for a double stroke cylinder according to the following cycle:
(a) Extend at (Tx10+U) l/min and 25 bar during (H+1) s; (b) No flow condition at 200 bar during 5 (H + 1) s; (c) Retract at (Tx10+U) l/min and 35 bar during 0.8(H + 1) s; (d) Stand still without flow at 200 bar to fulfil the cycle total time of 100 s. Answer the following:

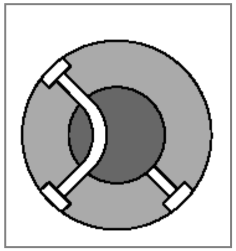
(1) Draw the flow rate and pressure evolution of the system with time.
(2) Obtain the surface ratio of the cylinder (relief valve at 250 bar) and the required oil volume by cycle. Compare it with the pumped oil.
(3) Power and efficiency of the cycle.
(4) An accumulator is added to enhance efficiency and the pump speed is controlled to deliver the required oil by cycle. Obtain its dimensions, that is obtain the required ϑ_{ac} if it operates between 200 and 250 bar.

Functions and standard denomination.

- A **directional control valve** lets the circuit to conduct the fluid to the different pipes, according to the phase needs, cycle requirements, opening or closing the ways.
- The Directional control valves are identified with two numbers, the first one indicates **the possible ports** and the second one indicates **the valve positions**.
- **Example: Valve with 2 ports and 2 positions.**







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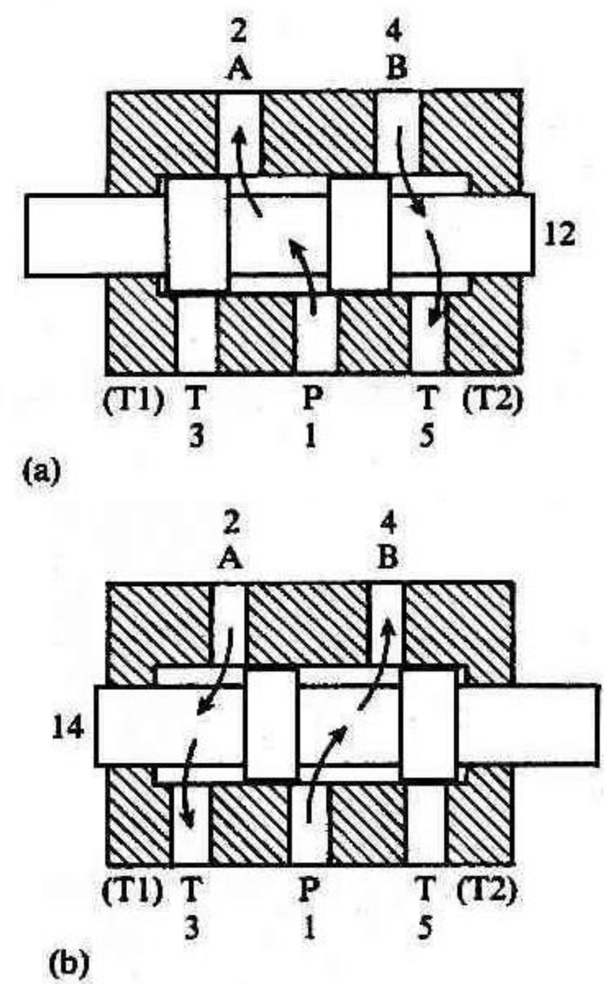
Classification and types.

- **Non-return (check) valves:**
  - Direct.
  - Piloted.
- **Directional control valves 2/2**
- **Directional control valves 3/2**
- **Directional control valves 4/2**
- **Directional control valves 4/3**
- **Directional control valves 5/2**
- **Directional control valves 5/3**

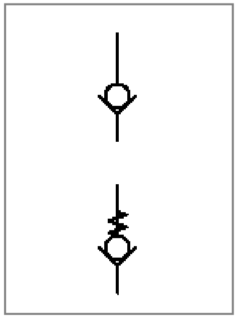
- **Example: Valve 5/2**



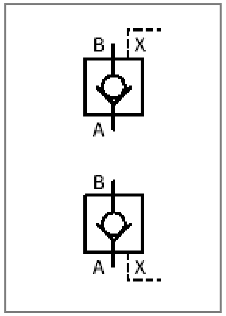
Non-return or check valves (I).

- Allow the fluid flow in one direction, blocking it in the other one.

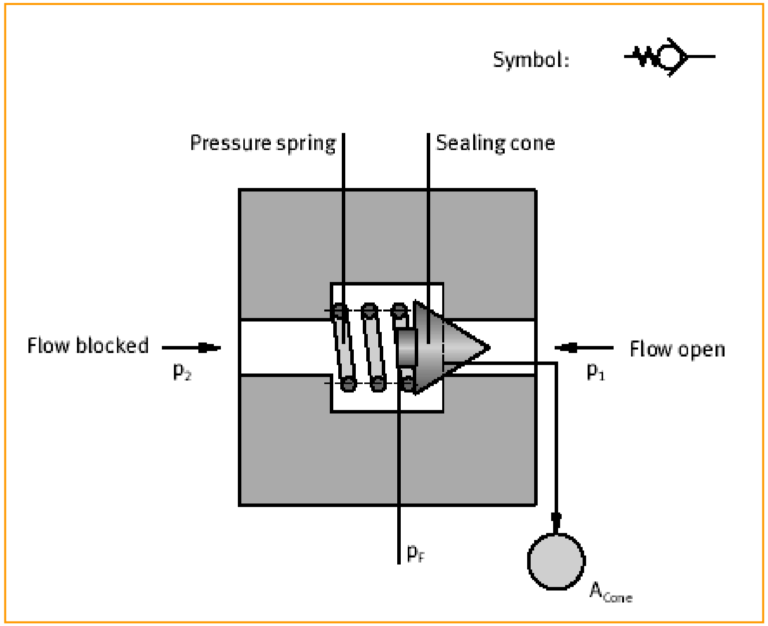
SIMPLE ONES
(With or without spring)

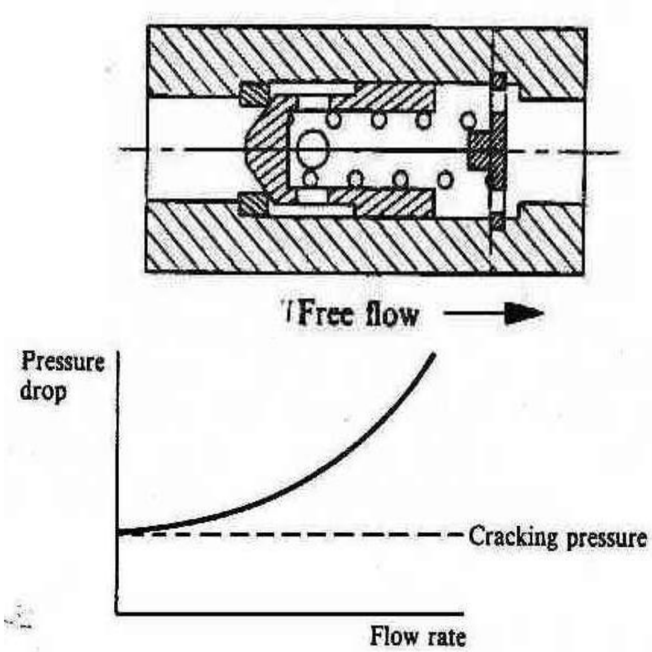


PILOTED (opening or closing side)



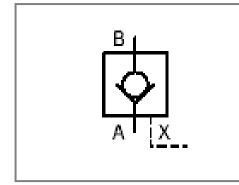
SIMPLE (or DIRECT)



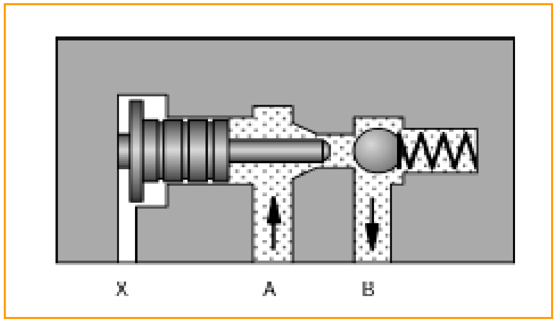


Non-return or check valves (II).

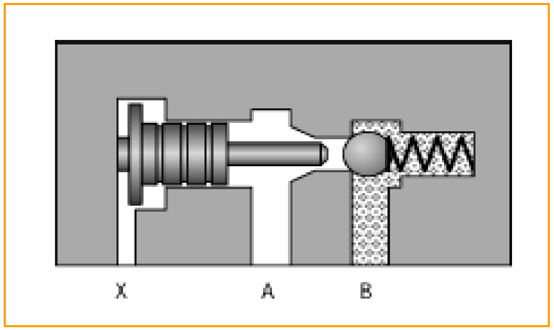
PILOT
OPENING



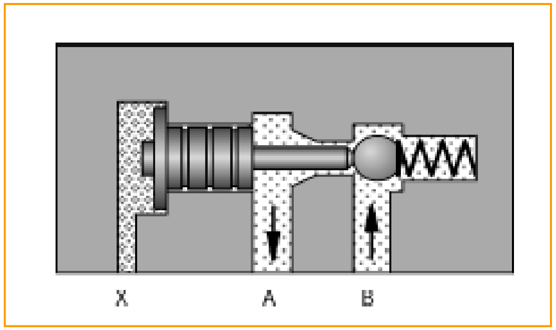
- **PILOT OPERATED (opening):** In one direction, they always allow the fluid flow. In the other one, they only allow the flow when the pilot pressure signal indicates so.



The flow goes from A to B in any case. (A small pressure opens the spring and moves the ball)

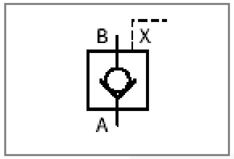


Without the signal X, the flow from B to A it is blocked, preventing the fluid to go in that direction. The spring also helps in that , unless…



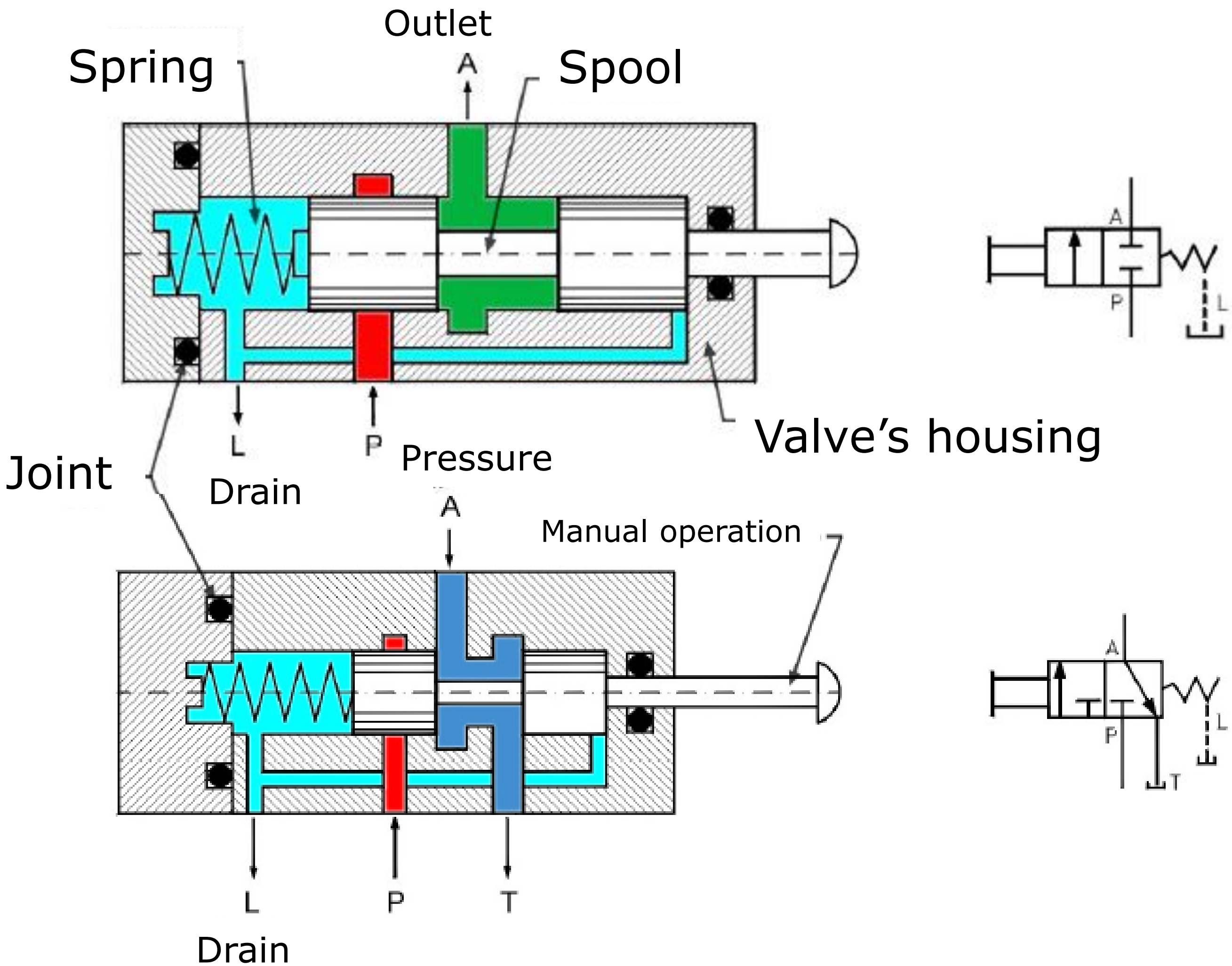
… the pilot signal in X becomes higher than the spring force, moving the spool and allowing the flow to go from B to A.

PILOT
CLOSURE

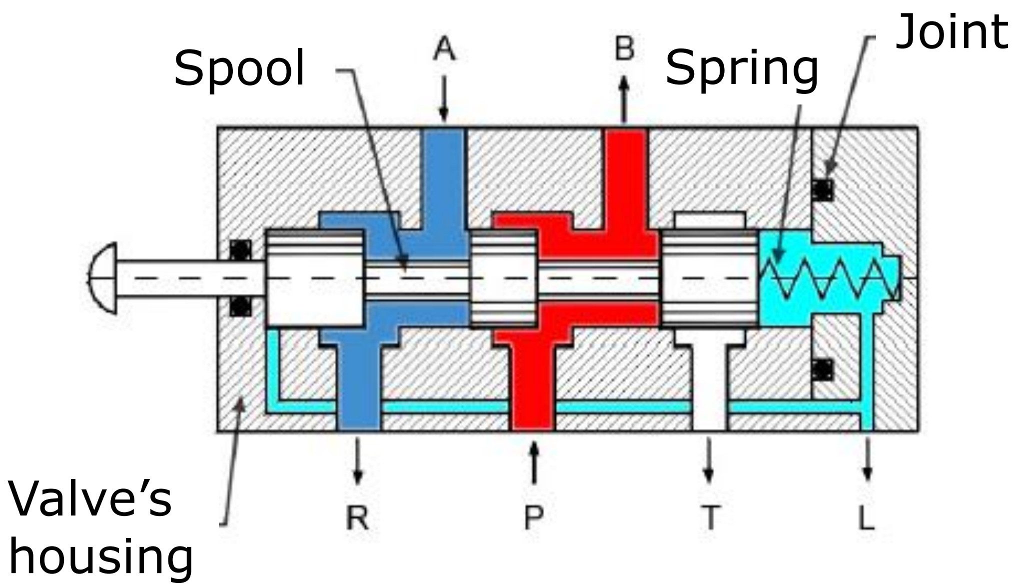


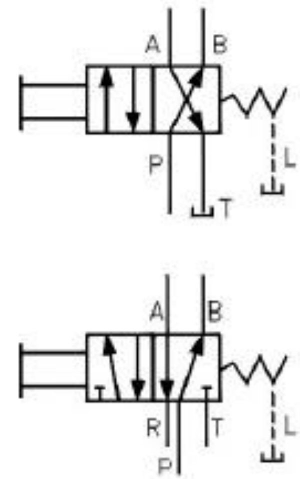
- **PILOT OPERATED (closure):** In one direction, they always block the fluid flow. In the other one, they might also close the valve, when there is a pressure signal in the port.

Directional control valves 2/2 and 3/2.

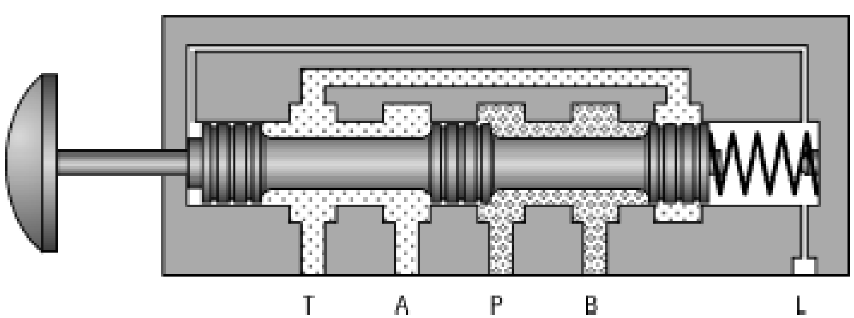


Directional control valves 4/2.

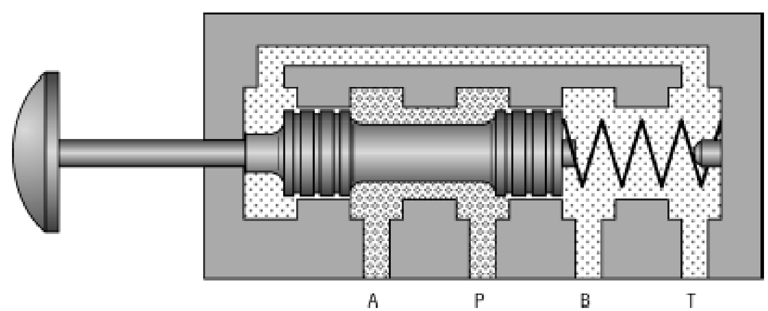




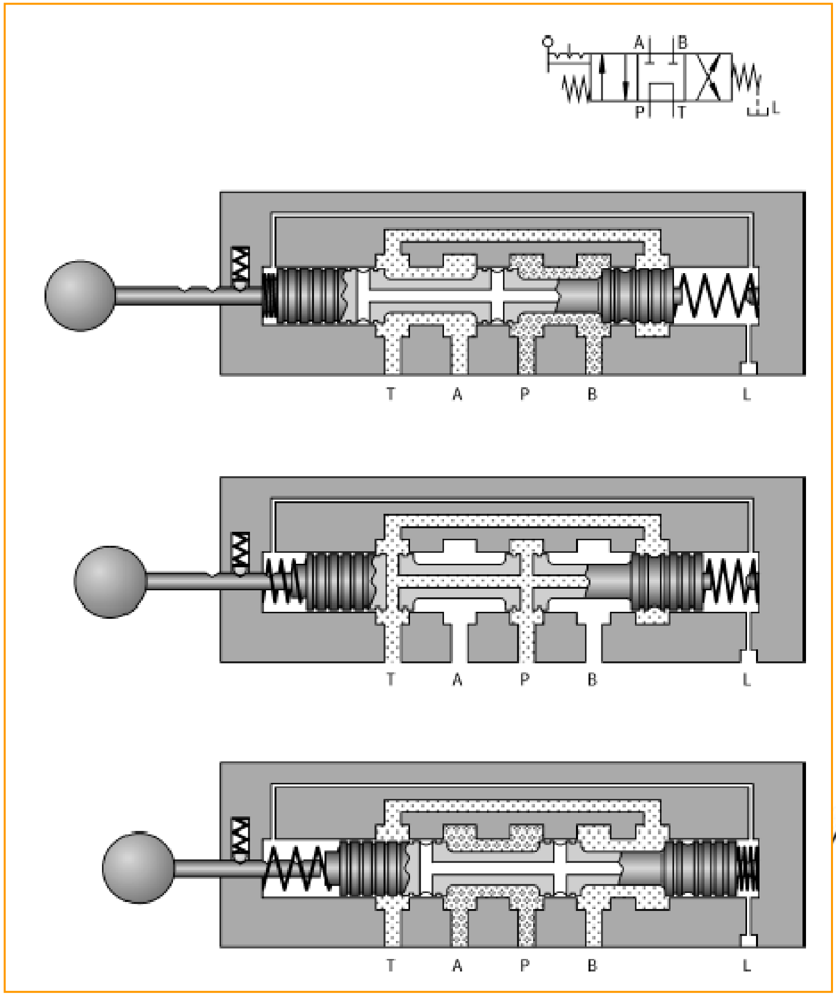
3 PISTONS

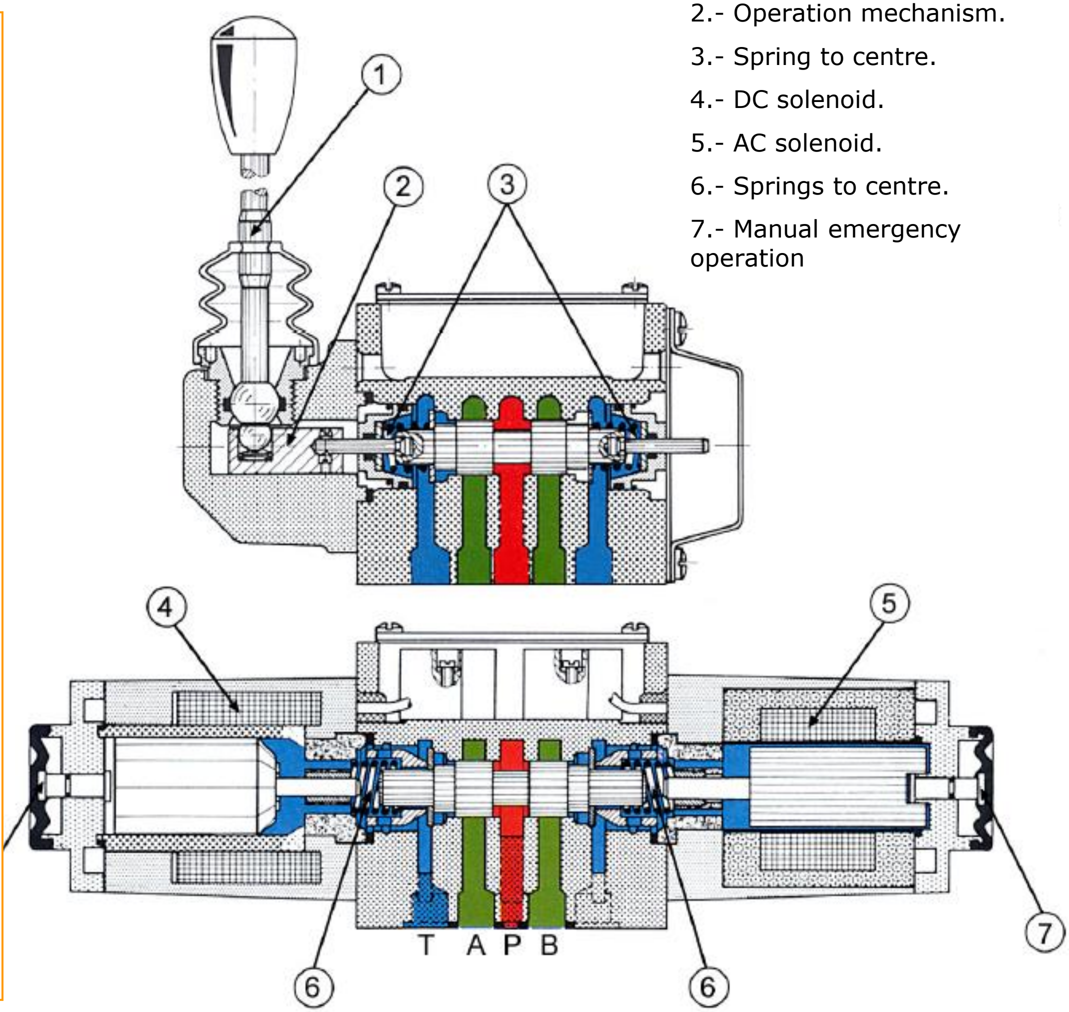


2 PISTONS (without drain port)



Directional control valves 4/3.





Legend:

1.- Lever.

2.- Operation mechanism.

3.- Spring to centre.

4.- DC solenoid.

5.- AC solenoid.

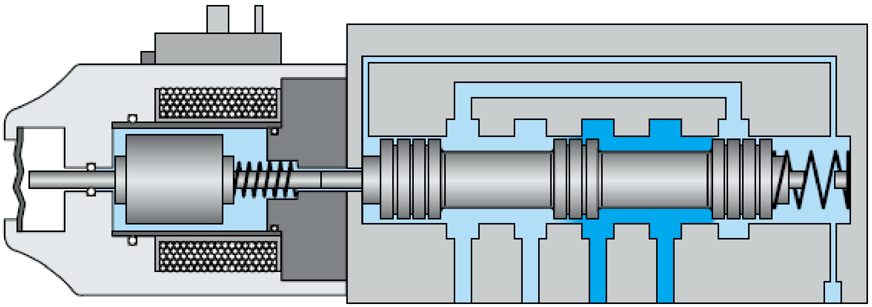
6.- Springs to centre.

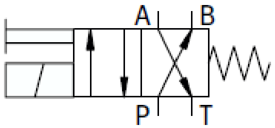
7.- Manual emergency operation

2.4. Directional control valves.

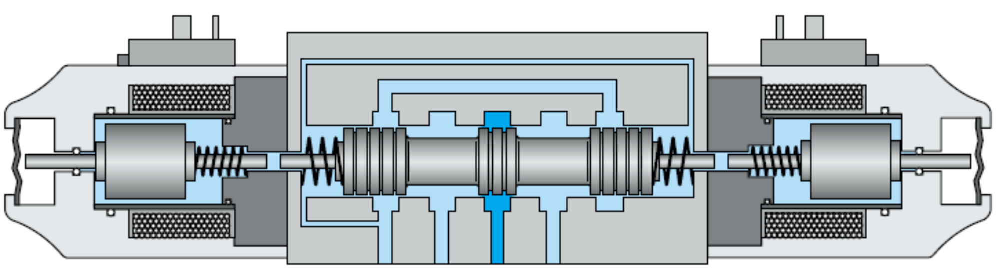
Directional electro-valves 4/2 and 4/3.

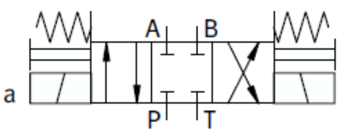
ELECTRO-VALVE 4/2





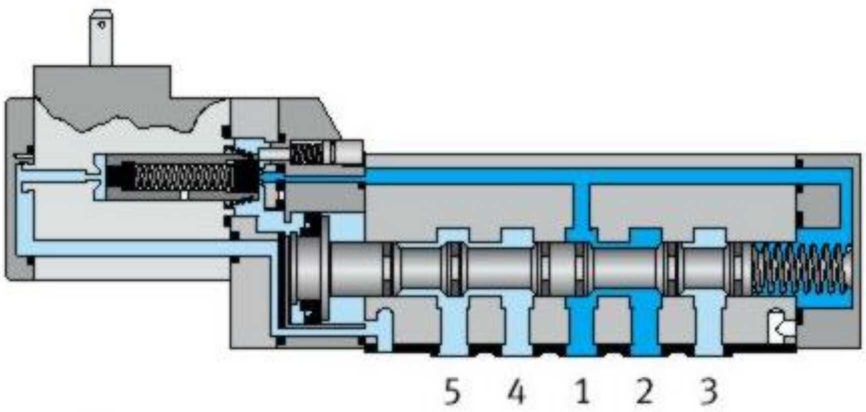
ELECTRO-VALVE 4/3

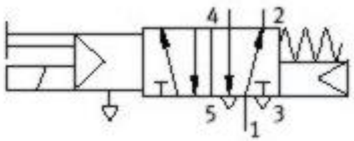


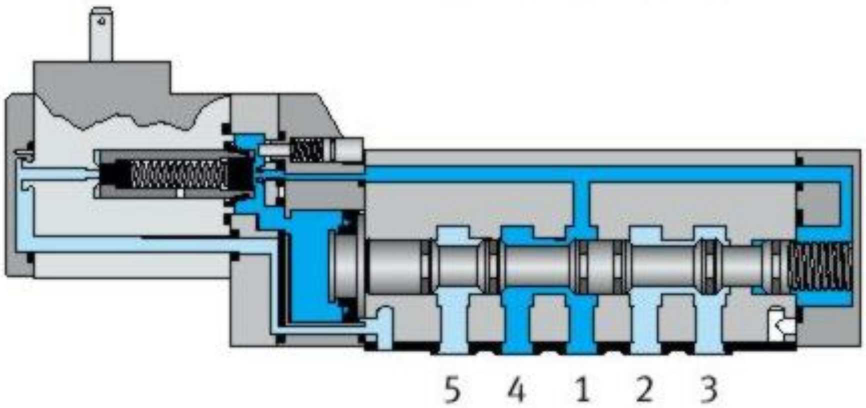


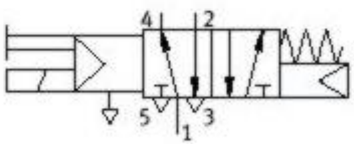
Directional control valves 5/2.

5/2 Electro-valves with servo control



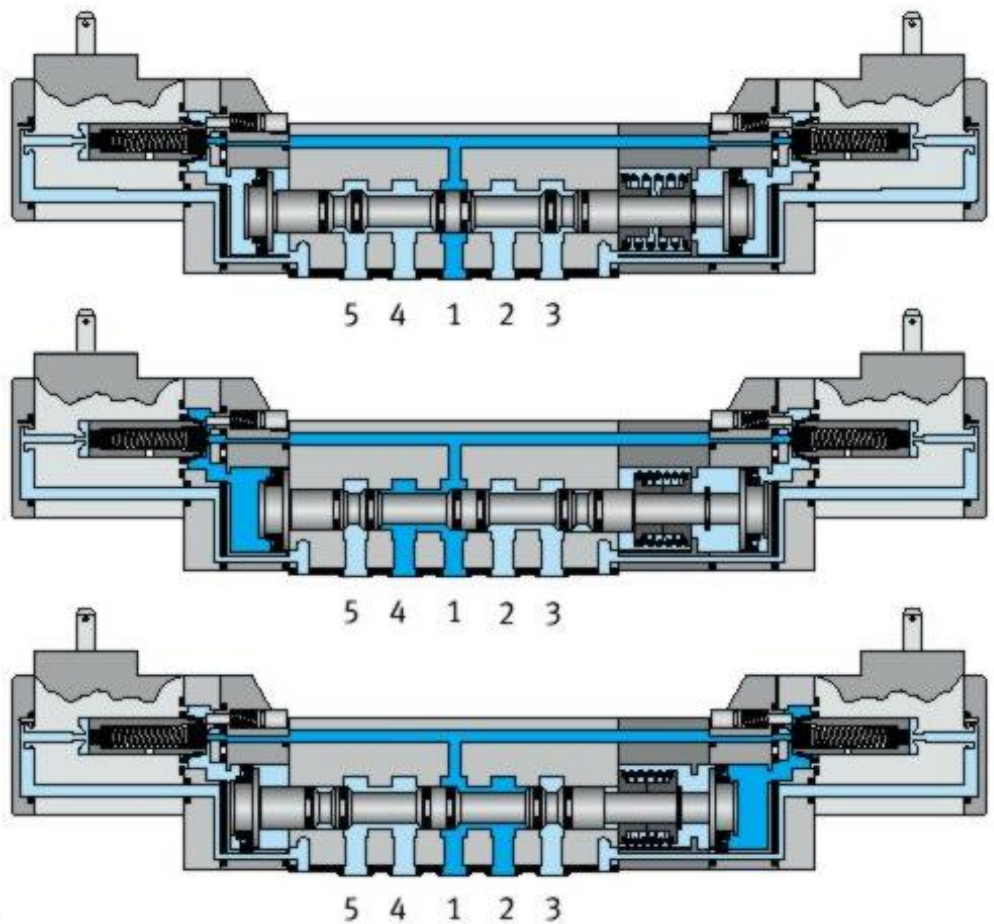


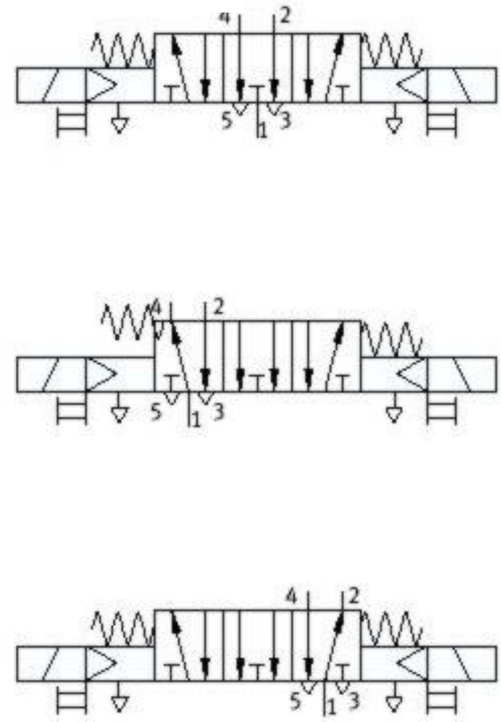




Directional control valves 5/3.

5/3 Electro-valves

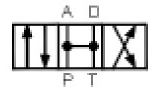




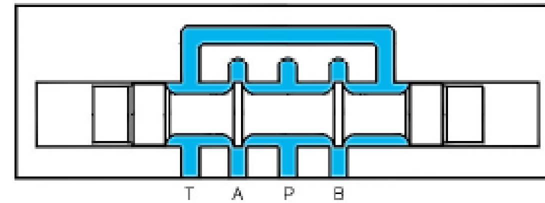
2.4. Directional control valves.

Types of centers (I).

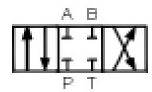
- **OPENED center.** All the oil coming from A and B returns to tank. The pump is connected also to tank and only the losses are to be considered.
- **CLOSED center.** The pump discharges to tank trough the PLV. A and B remain isolated and then there will be oil in both lines, one of them with pressure.
- **FLOATING center.** The pump discharges through the PLV. Now, A and B are connected to tank.
- **H center.** The pump discharges to tank. Line B remains isolated.
- **TANDEM center.** The pump is connected to tank and works without pressure. Lines A and B are blocked, with no movement allowed in both of them.



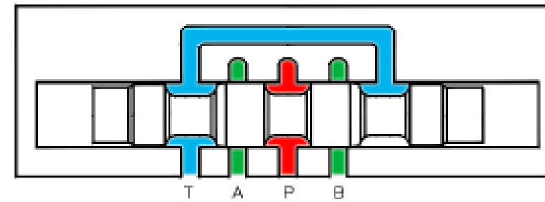
Opened center



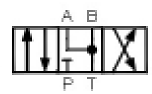
All connected



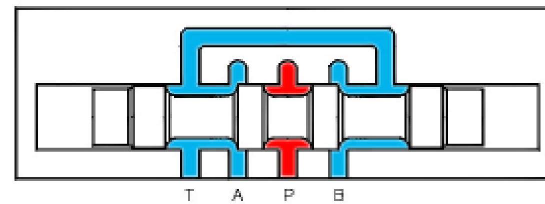
Closed center



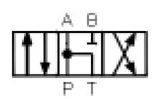
All closed



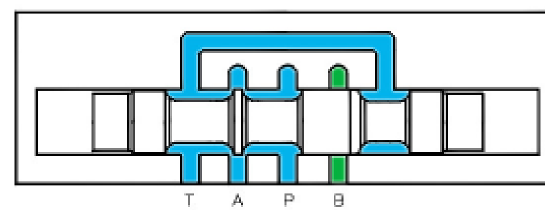
Floating center



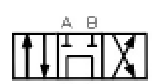
P closed, A and B to tank



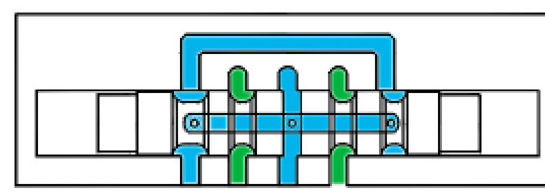
H-center



B closed, A and P to tank



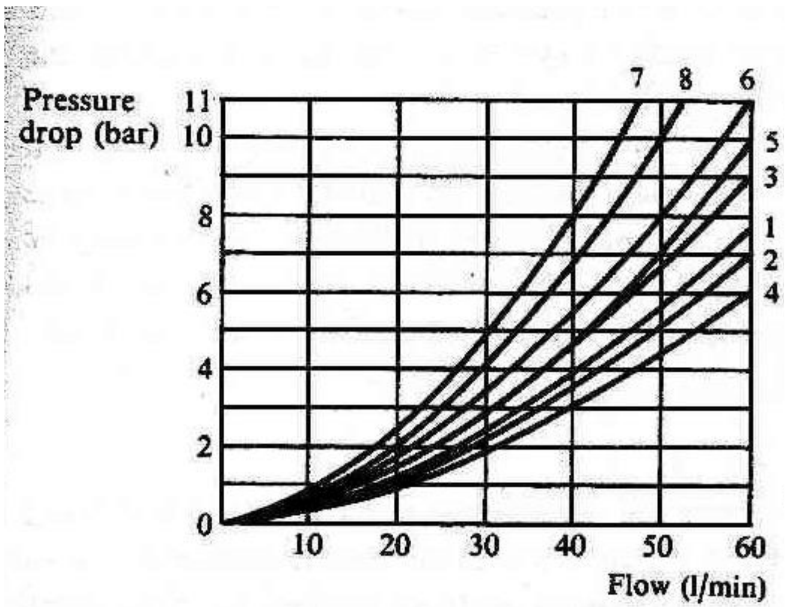
Tandem center



A and B closed, P to tank

Types of centers (II) and head losses.

Spool reference	Center condition	Switching characteristic or typical application
a		Prevents collapse of pressure during changeover (may cause pressure shocks)
b		Pump unloading (In two-position valve pressure collapses momentarily during change-over)
c		Unloading pump circuit but blocking ports A and B giving a degree of locking. (NOTE This spool causes a higher pressure drop through the valve than most other spools)
d		Pilot-operated check valve circuits. Hydrostatic transmission to give free-wheeling effect and reduce pressure surges. Used when a second directional valve has to be supplied with fluid
e		Single acting cylinder circuits
f		To gradually relieve pressure in the service lines on change-over to mid-position
g		To maintain pressure on both service ports in mid-position, e.g. clamping

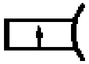
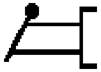
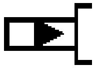
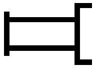
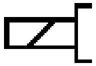
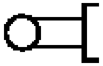
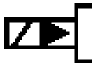
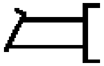
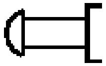
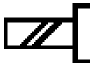


Spool type	Flow direction			
	P–A	P–B	A–T	B–T
a	3	3	1	1
b	2	4	2	2
c	5	3	6	6
d	1	1	2	1
e	3	1	3	3
f	1	1	2	1
g	2	4	3	3

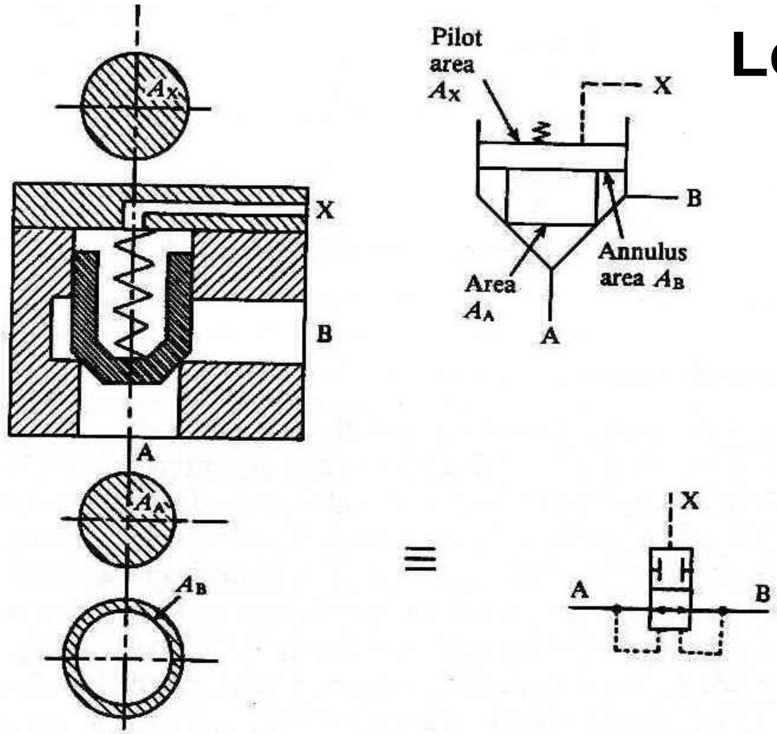
2.4. Directional control valves.

Types of valve operation systems:

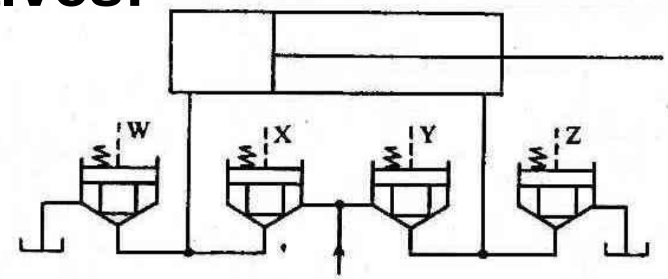
- Manual.
- Mechanical.
- Pneumatic.
- Electrical.
- Hydraulic.

Operation methods		Operation methods	
Pressure compensated		Lever	
Mechanical with detention		Pilot pressure	
Manual		Solenoid direct	
Mechanical - roller		Solenoid pilot	
Pedal		Spring	
Push button - Knob		Double solenoid	

2.4. Directional control valves.



Logic valves:



Equivalent spool valve state												
W	0	1	1	0	1	0	0	1	1	1	0	1
X	0	1	0	1	0	1	0	1	0	1	1	1
Y	0	1	1	0	0	1	1	0	1	0	1	1
Z	0	1	0	1	1	0	1	0	1	1	1	0

EXAMPLE: Consider a valve with $A_A{:}A_X = 1{:}1.1$ and $A_B = 0.1\ A_A$. The force exerted by the spring is equivalent to 3 bar and the pilot pressure is 7 bar.

When the flow goes from A to B, the required pressure in A to open the valve it is obtained by the forces equilibrium on the spool, that is:

$P_A\ A_A = (P_X + P_S)\ A_X \rightarrow P_A = (P_X + P_S)\ A_X/A_A = (7 + 3)\ 1.1 = 11$ bar

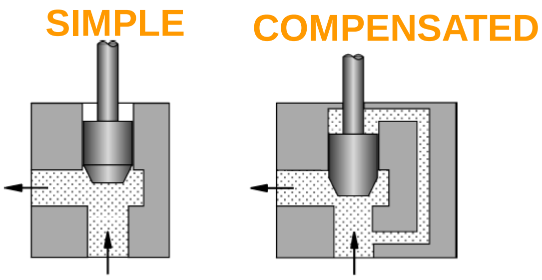
When the flow goes from B to A, the pressure required in B to open the valve is again obtained from the forces equilibrium on the spool, that is:

$P_B\ A_B = (P_X + P_S)\ A_X \rightarrow P_B = (P_X + P_S)\ A_X/A_B = (7 + 3)\ 1.1/0.1 = 110$ bar

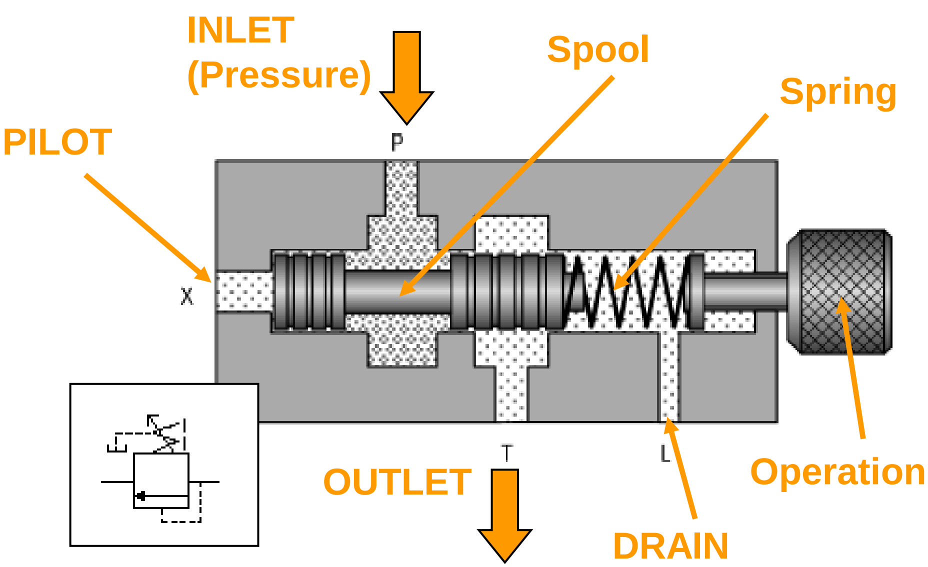
Function and basic elements.

- The **pressure control valves** are placed to limit and regulate the pressure in the different parts of a hydraulic system.

- Working principle:
  simple vs compensated



- **Constructive elements:**



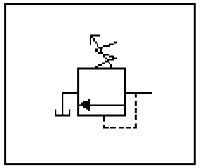
- **BODY,** with an internal sliding element (**SPOOL OR POPPET**), that closes the flow path.

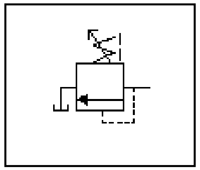
- The position of the spool depends on the relative value of the **FLUID PRESSURE** and the **ELASTIC SPRING FORCE**.

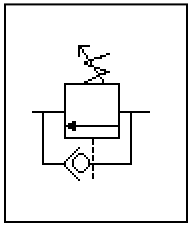
- Two ports, namely INLET and OUTLET, and also **PILOT** and **DRAIN** (internal or external leaks).

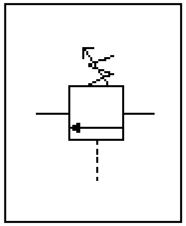
Classification:

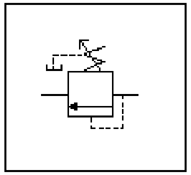
- **Relief Valves (RV)**
  - Direct.
  - Pilot operated.
- **Load compensation valves**
- **Discharge valves**
- **Sequence valves**
  - Direct.
  - Pilot operated.
- **Pressure reduction valves.**
  - Direct.
  - Pilot operated.

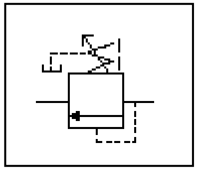


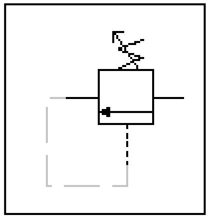






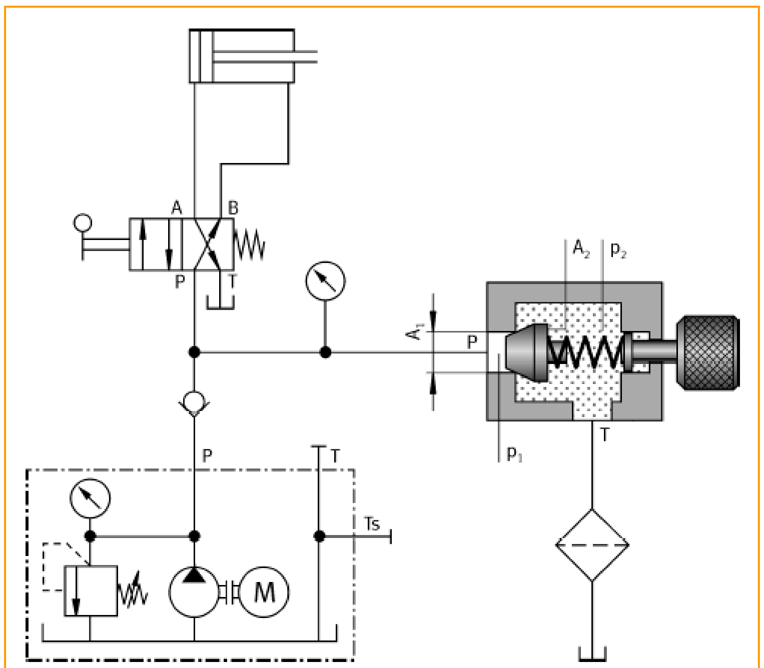




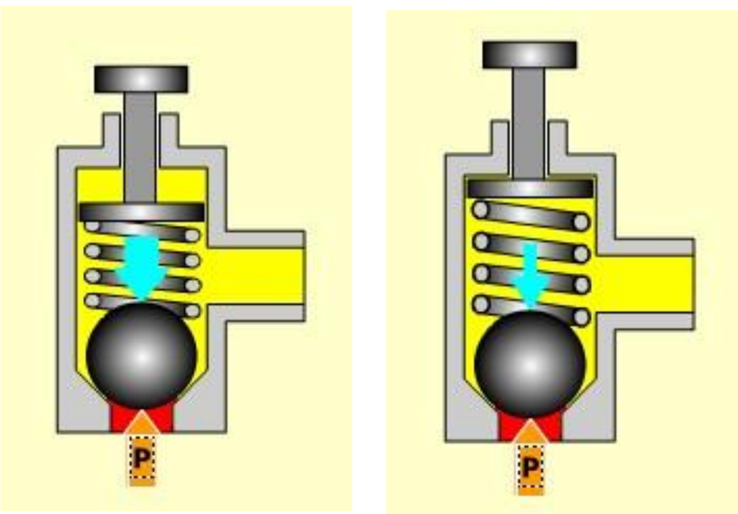


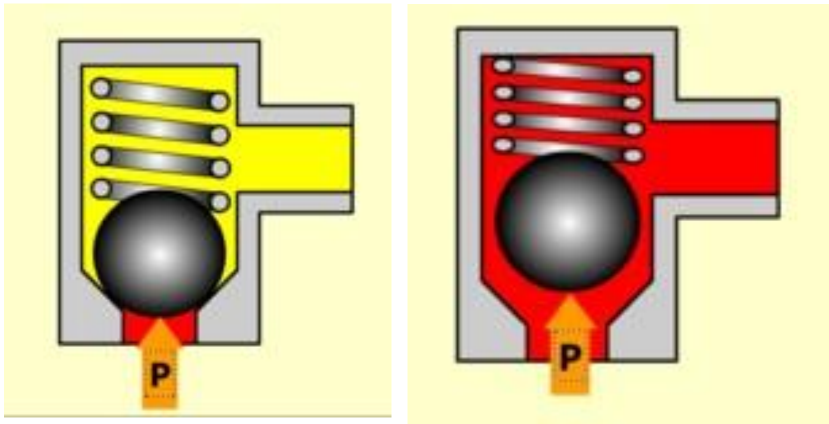
Pressure relief valves (I).

- **Relief valves.** Protect the pump against an overpressure in the system, limiting the maximum possible pressure.
- **Direct or pilot actuated.**



DIRECT





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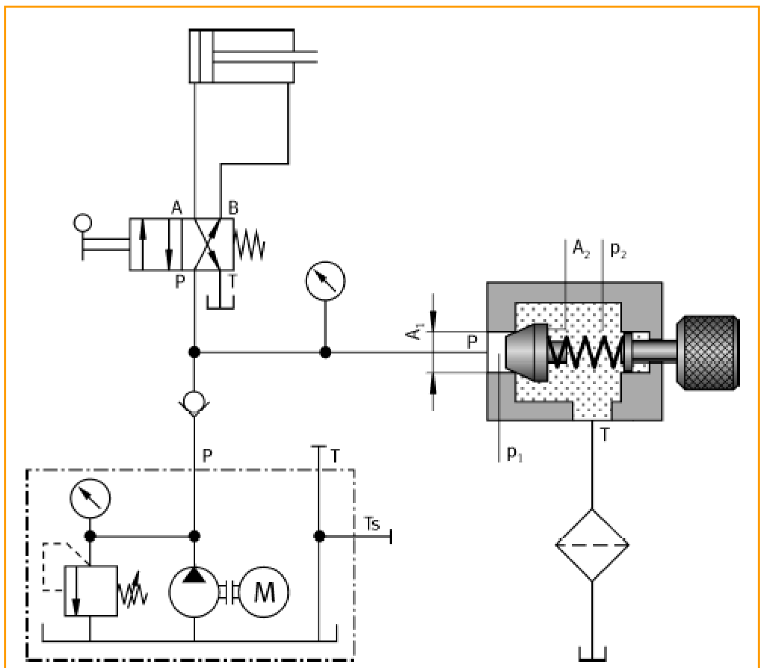
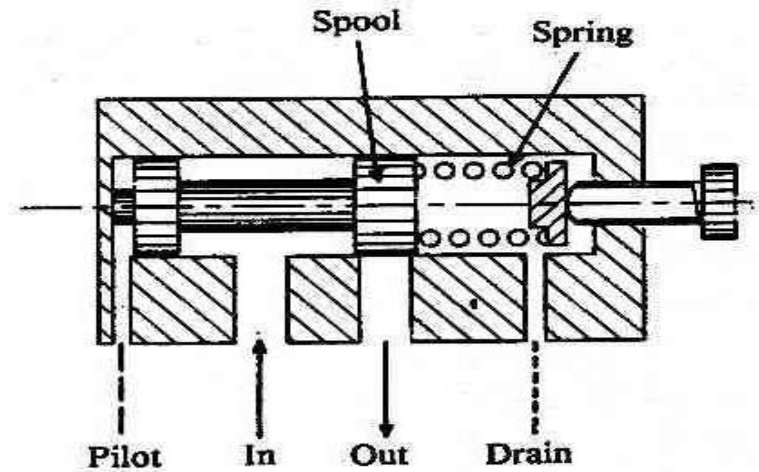
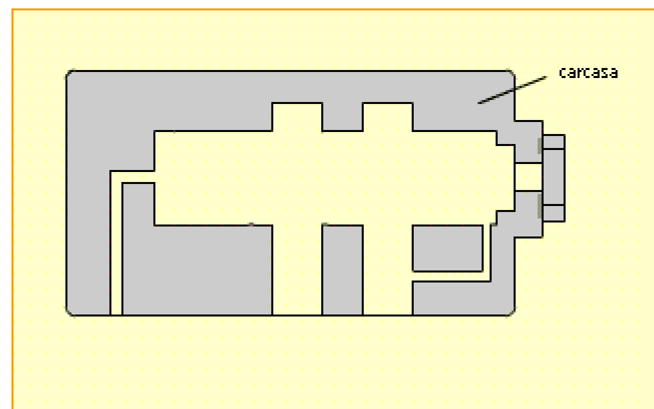
Pressure relief valves (II).

- **Relief valves.** Protect the pump against an overpressure in the system, limiting the maximum possible pressure.
- **Direct or pilot actuated.**



PILOT ACTUATED





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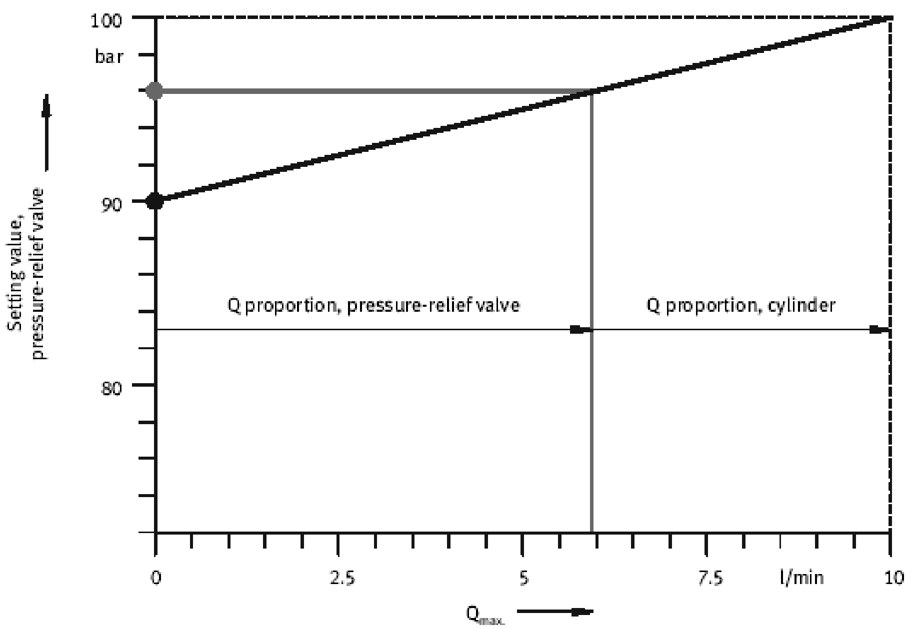
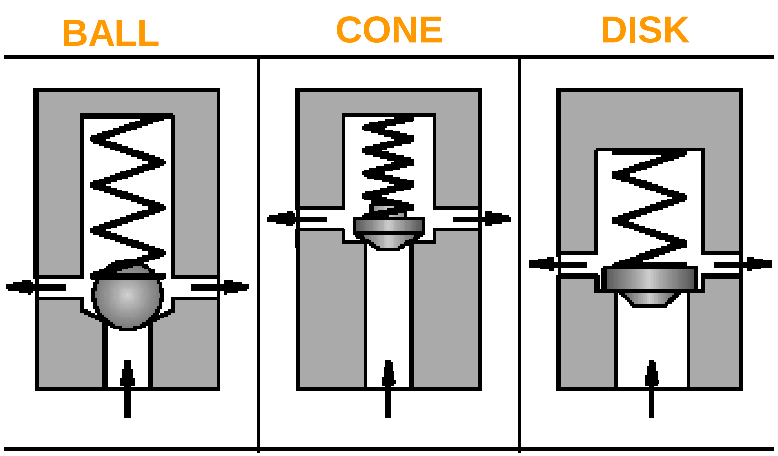
Pressure relief valves (III).

- **OVER PRESSURE MARGIN:** Difference between the minimum pressure to open the valve and the pressure drop through it when the maximum allowed flow rate passes by it.
- Needs **FILTRATION.**
- **HYSTERESIS:** Difference on opening and closing pressure values.
- **TRANSIENT:**
  - Water hammer.
  - Tend to whine.
  - Vibrations.
- **PARALLEL CONFIGURATION.**

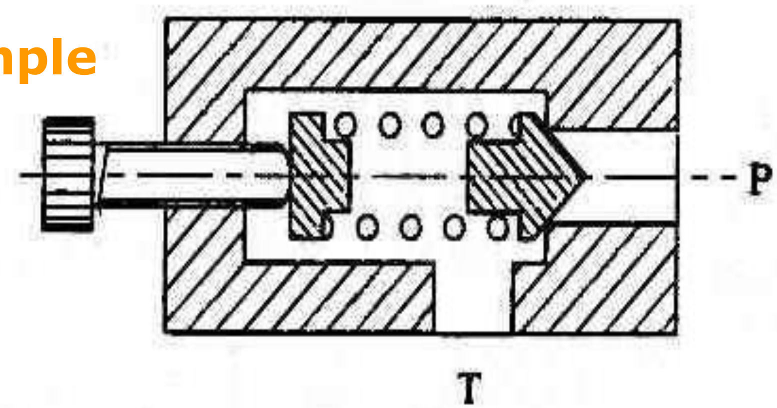


Pressure relief valves (IV) – Types of seats.

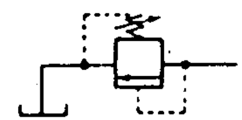
- **Simple**.
  - Cone or ball (Whine, seat damage. Well fitted for infrequent duty).
  - Guided piston (100 bar). Well fitted for continuous duty.
  - Differential (350 bar).
- **Double stage.**

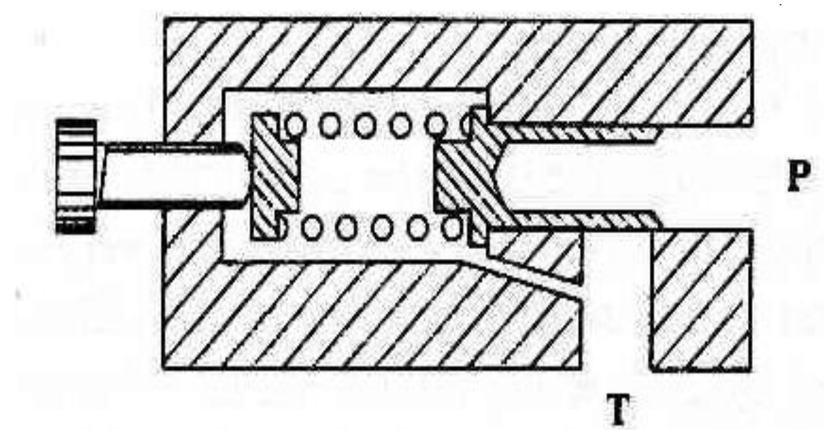


Simple



Guided piston





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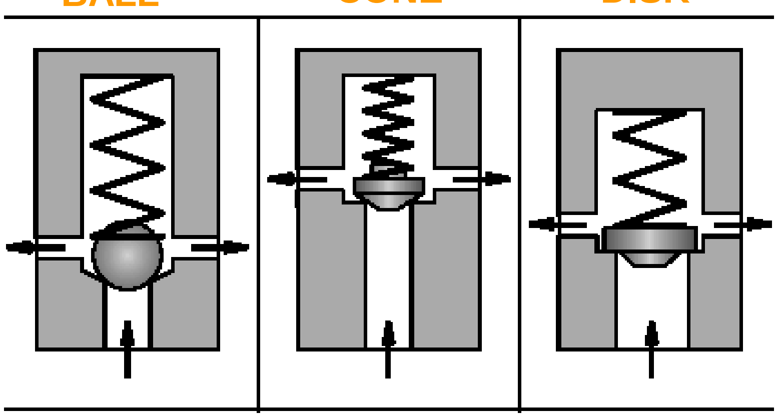
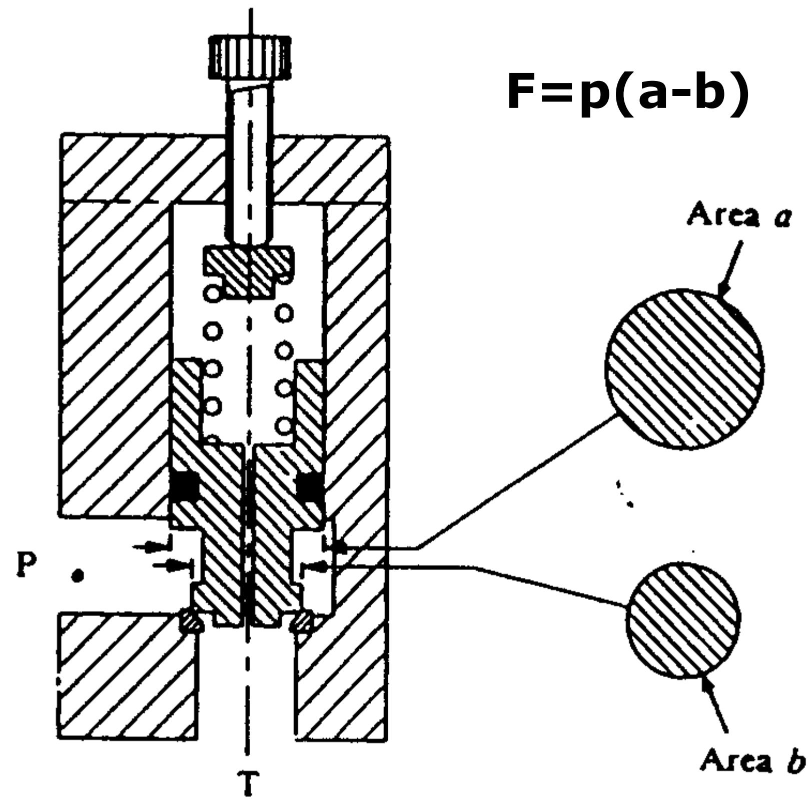
  - Differential (350 bar).
- **Double stage.**

BALL CONE DISK



Differential

F=p(a-b)



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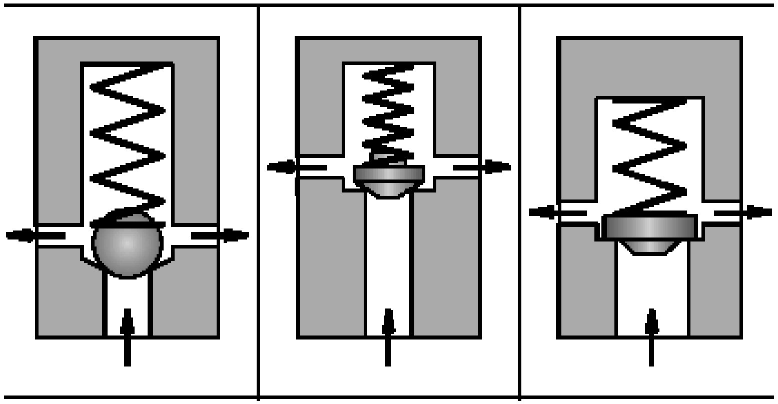
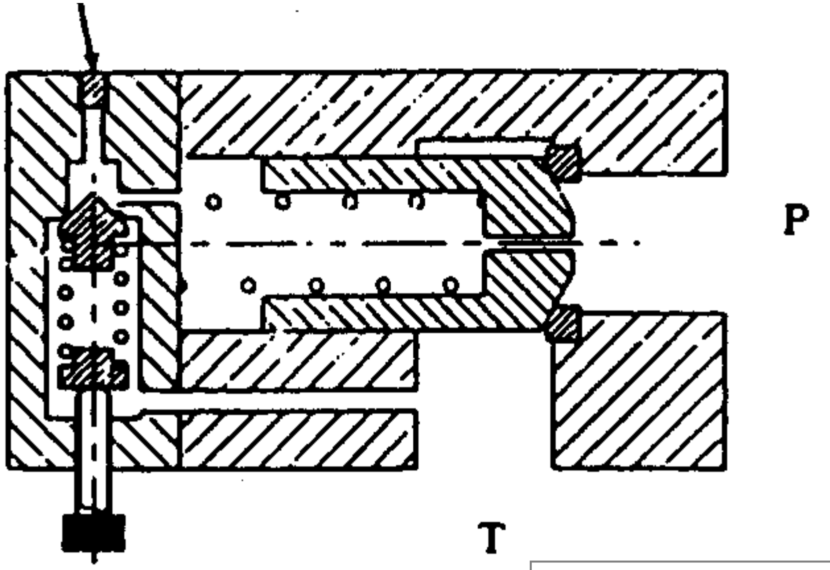
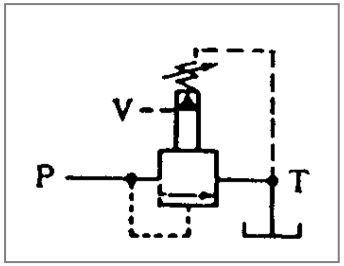


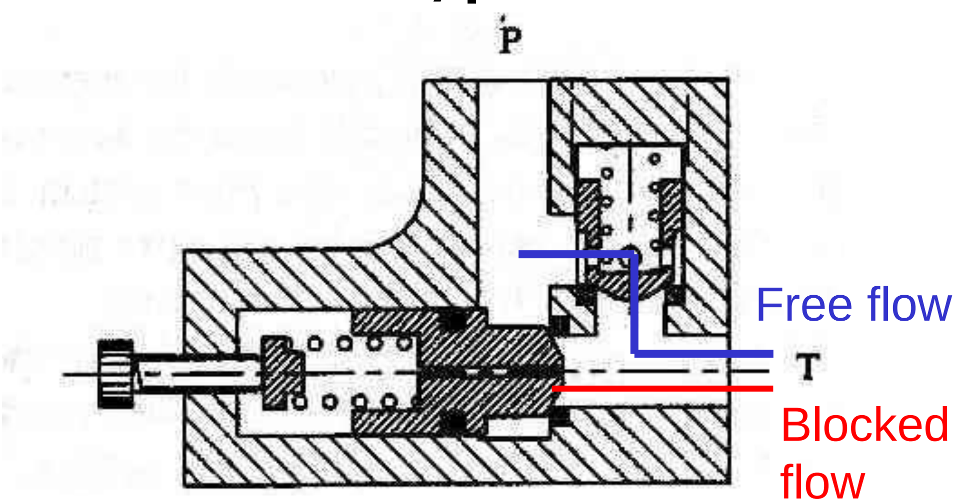
Double stage

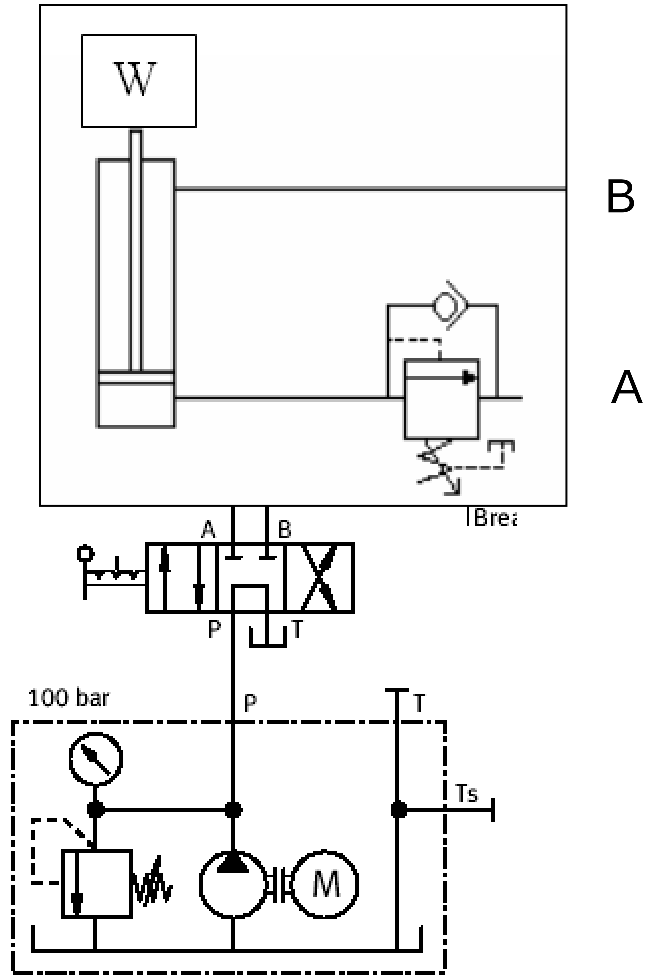




Pressure valves as counterbalance valves (I).

- Classical application: avoid the **speeding up of a load.**
- **Extending**: Free flow through the check valve.
- When **retracting**: The flow must open the pressure valve.
- **Pressure drop**: the load is kept in its position.
- Possible **in-line/parallel**

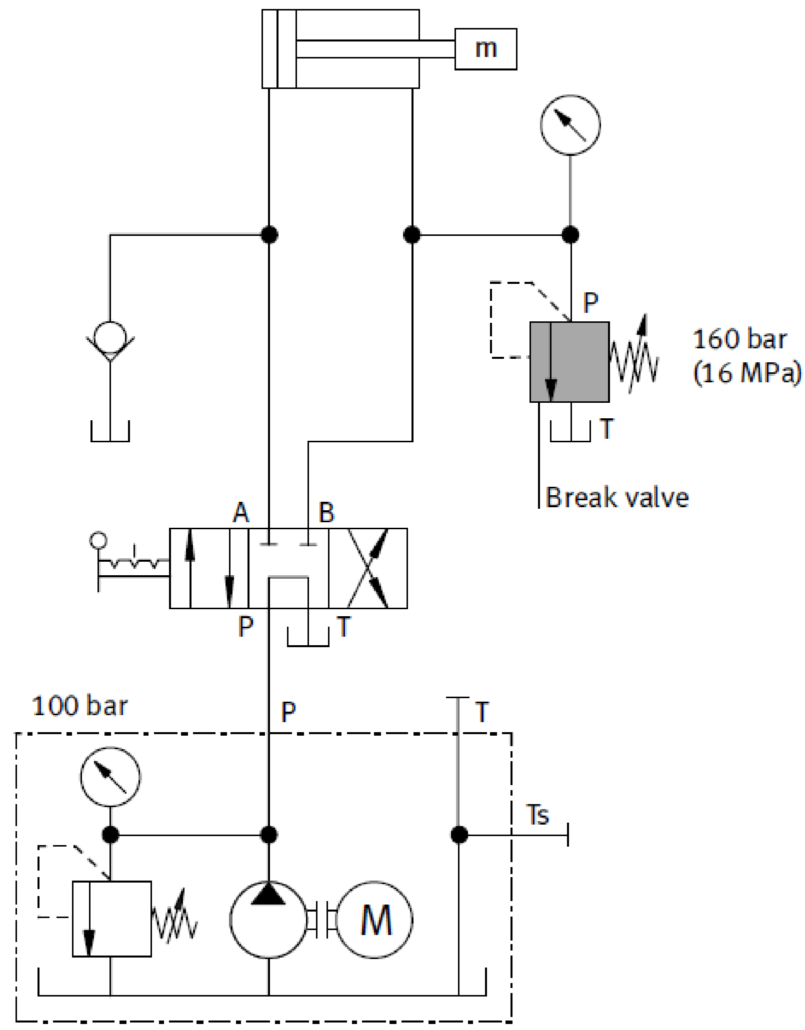




Other use: braking valve.

In this use, the valve acts to avoid **pressure peaks** that can occur as a consequence of the **inertia** of the loads when a directional valve is suddenly closed.

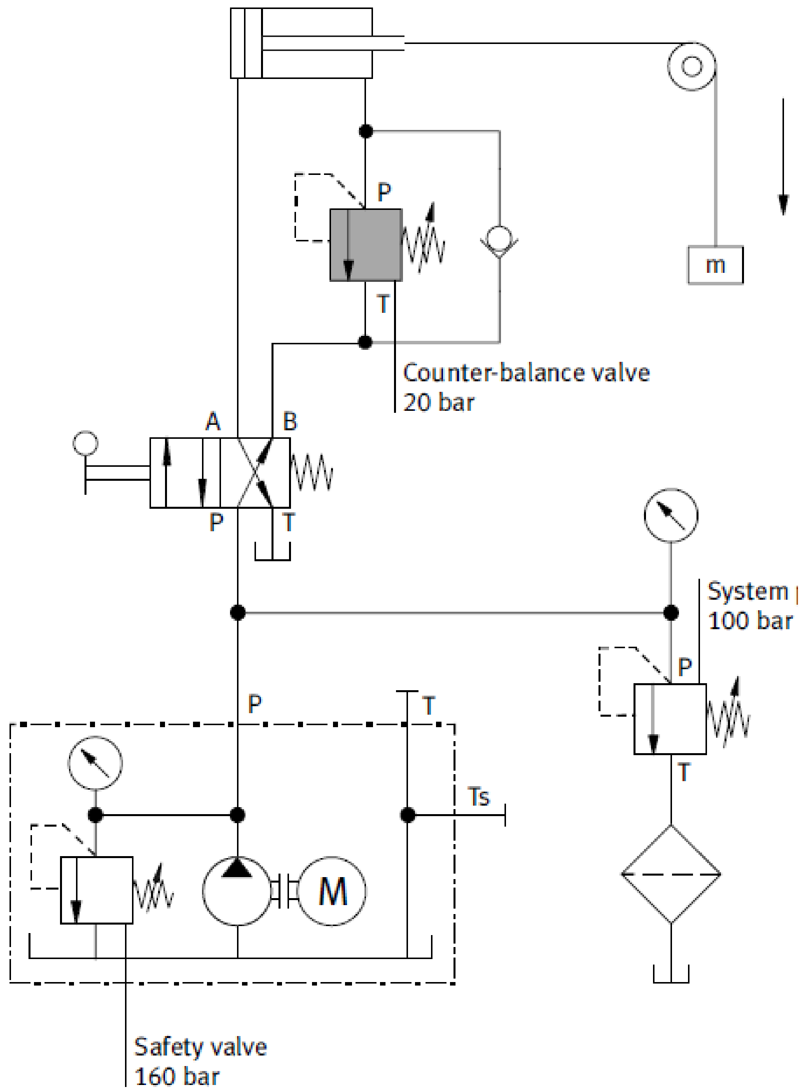
Exercise: A mass, m = 80000 kg, it is moved in the extent stroke in the figure with a speed of 1 cm/s. Determine the stopping time when the directional valve is placed in its central position. It is known that the pump delivers a flow rate of 3 l/min when extending and that the surface quotient in the cylinder is 1.25. The friction coefficient with the surface is μ=0.4



Exercise 5: UO: UOxxxHTU

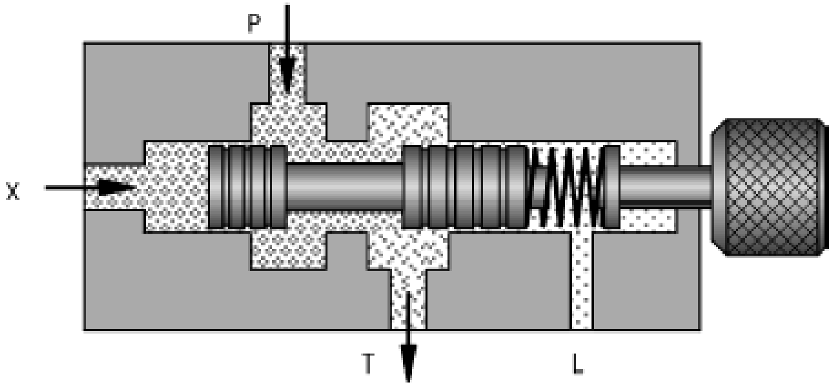
A given cylinder with bore section (40 + 5xU) cm^2 is being used with a pump with (H + 1) l/min. To avoid the over-run of a load with mass m, a compensation valve set at 20 bar is used. If the total stroke is 25 cm and the retract time is (0.5xT + 1) s, determine:

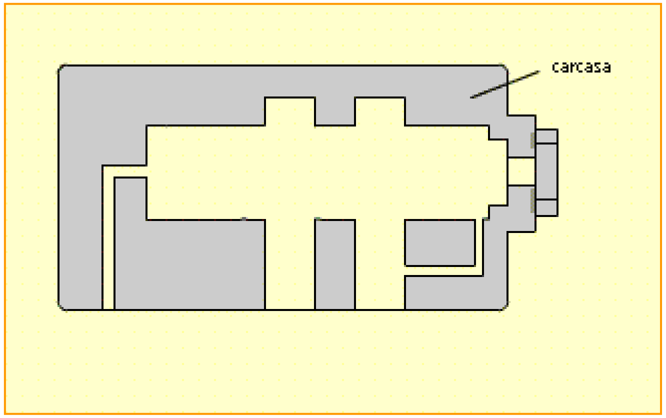
a) The cylinder surface quotient.
b) The maximum value of the load that might be used with the current valve. What is the pressure in the rod side of the cylinder, for this maximum value?
c) Required power to retract the load.
d) Final speed and acceleration when reaching the floor if the load is two times the maximum one.



Unloader valves.

- A relief valve which is unloaded to tank when it RECIVES an opening signal.
- Normally with a pilot from the pressure in another place in the system (**external pilot**).
- **PARALLEL configuration**.





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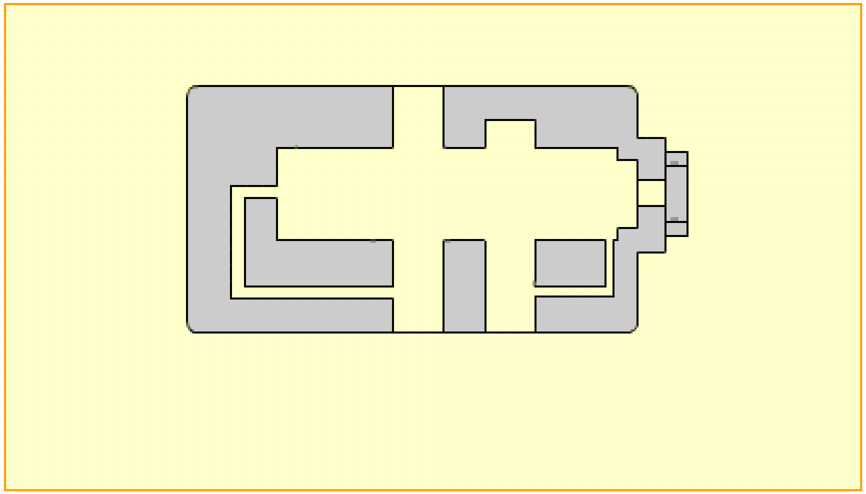
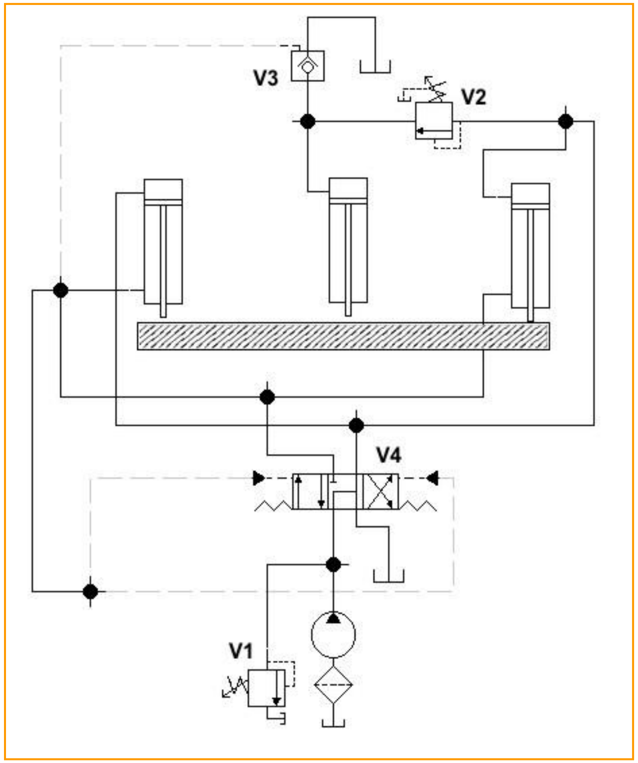
Sequence valves.

- They allow the flow to go into a given part of the circuit when the opening pressure is reached.
- External or internal pilot pressure. External drain.
- **IN LINE**.





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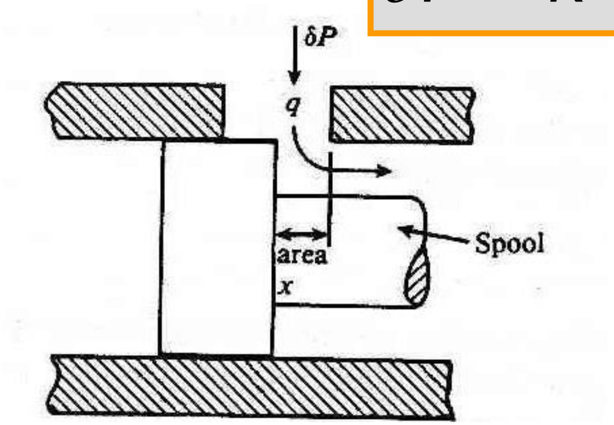
Functions and operation.

- The **flow control valves** are placed to regulate and control the delivered flow rate in the different parts of a hydraulic system.
- Quite often, they are built together with a **relief valve**, that directions the flow towards the tank (except in the pressure compensated valves).
- The flow rate **is controlled changing the cross section or area** A(x), using an obstruction, and producing a subsequent head loss:

$$\boxed{\delta P = k'(x)\ \ q^2} \Rightarrow \boxed{q = k(x)\,(\delta P)^{1/2}}$$

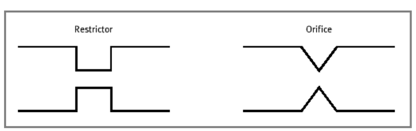


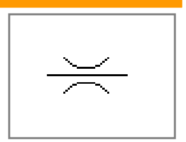
- The control can be **simple o compensated.**

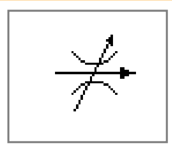
Classification: Valves types.

FLOW REGULATION VALVES

- **SIMPLE REGULATION**
  - **Orifice type**
  - **Flow restriction**
- **REGULATION AND CONTROL**
  - **Compensated**







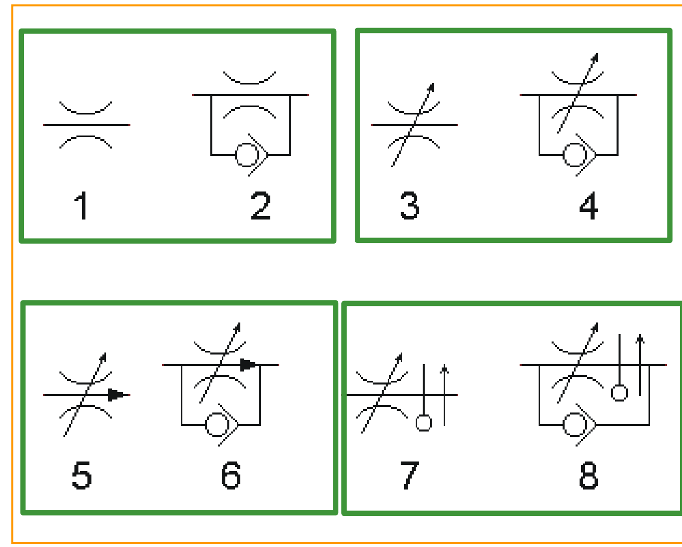
$$Q_{regulated} \neq \text{constant}$$

Resistance to the flow passing. Load dependent.

$$Q_{regulated} = \text{constant}$$

Load independent.

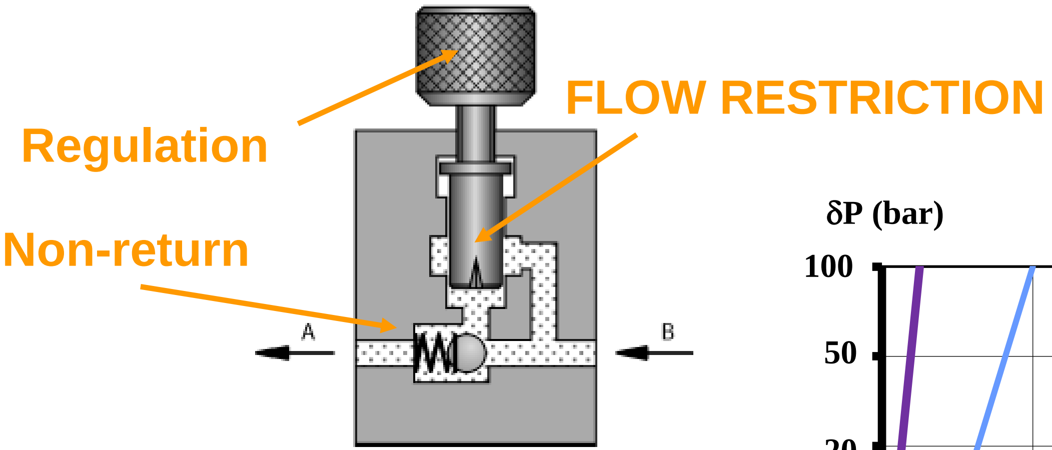
Simple **Adjustable**

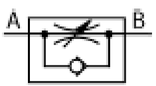


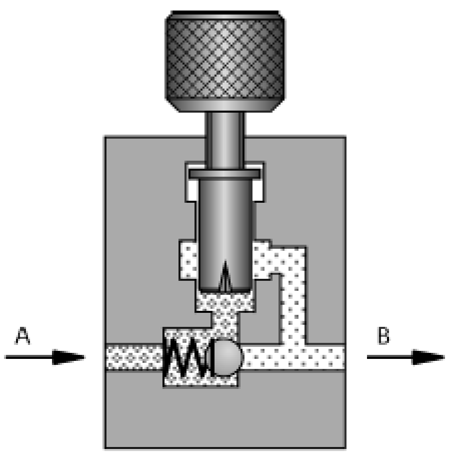
Pressure Compensated **Temperature Compensated**

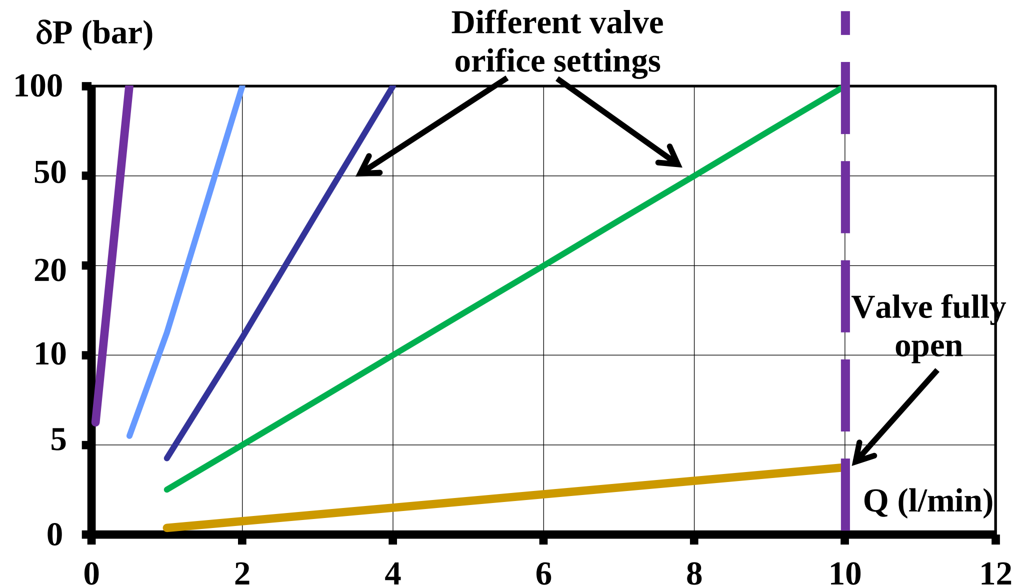
Basic elements: head loss.

MIND: In this particular operation, the behaviour is quite similar to the laminar flow (linear variation with flow rate)









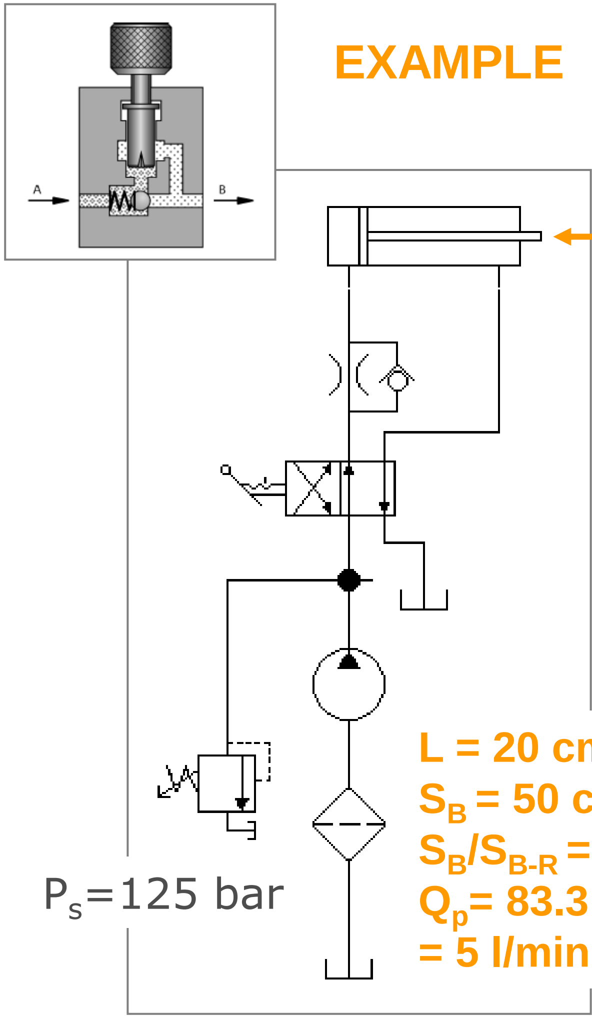
Different values for a simple needle valve

Problem: If the LOAD changes, the flow rate changes

2.6. Flow control valves.

Simple control valves (I).

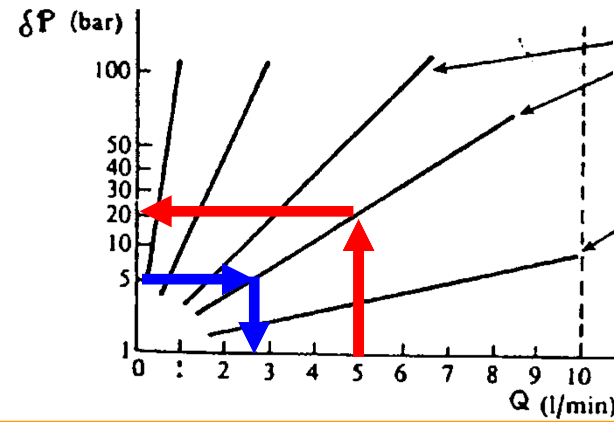
EXAMPLE



F = 50 kN.

L = 20 cm
S_B = 50 cm²
S_B/S_{B-R} = 2
Q_p= 83.3 cm³/s = 5 l/min.

Different values for a single needle valve



F = 60 kN.

$$P = \frac{F}{A_B} = \frac{60000\,\text{N}}{50\,\text{cm}^2} = 120\,\text{bar}$$

As: $Q_P = 5$ l/min and $\Delta P_V = 20$ bar, $P_P = 140$ bar > 125 bar.

Therefore, the valve has opened before. What is the flow rate division?

Fixing: $P_P = 125$ bar and P = 120 bar, therefore $\Delta P_V = 5$ bar and $Q_E = 2.5$ l/min

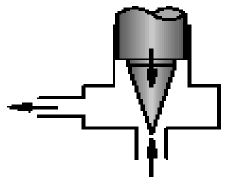
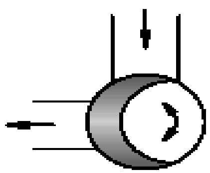
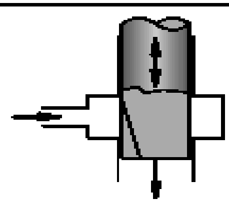
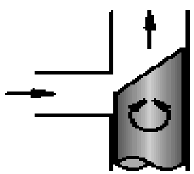
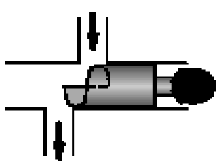
And, then, $t_E = \frac{\vartheta_B}{Q_E} = \frac{1\text{l}}{2.5\,\text{l/min}}$ that is, $t_E = 24$ s.

Problem: If LOAD changes, then the FLOW RATE and the CYCLE TIME change.

2.6. Flow control valves.

Simple control valves (II).

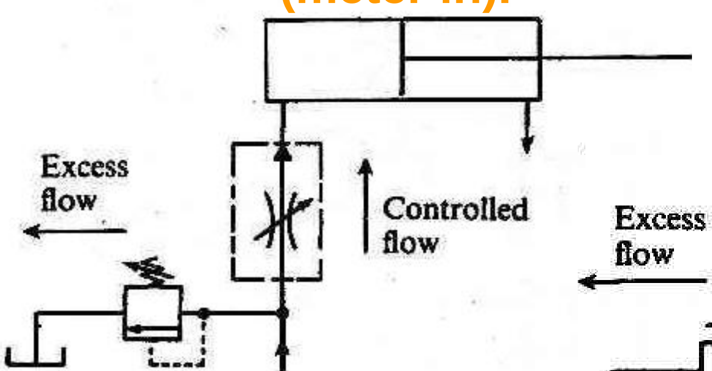
Types of flow restriction

Type		Resistance	Dependence on viscosity	Ease of adjustment	Design
	Needle restrictor	Increase in velocity, high friction owing to long throttling path	Considerable owing to high friction	Excessive cross-sectional enlargement with a short adjustment travel, unfavourable ratio area to control surface	Economical, simple design
	Circum-ferential restrictor	As above	As above, but lower than for the needle restrictor	Steadier cross-sectional enlargement, even ratio area to control surface, total adjustment travel only 90°.	Economical, simple design, more complicated than the needle restrictor
	Longitudinal restrictor	As above	As above	As above, however sensitive adjustment owing to long adjustment travel	As for circumferential restrictor
	Gap restrictor	Main part: increase in velocity, low friction, short throttling path	Low	Unfavourable, even cross-sectional enlargement, adjustment travel of 180°	Economical
	Gap restrictor with helix	Increase in velocity, maximum friction	Independent	Sensitive, even cross-sectional enlarge-ment, adjustment travel of 360°	Expensive to produce helix

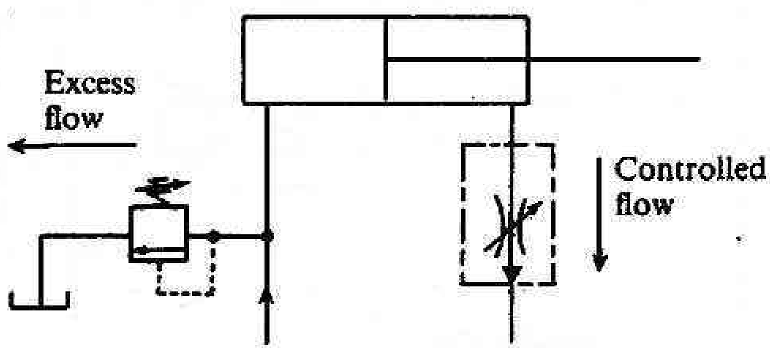
Simple control valves (III).

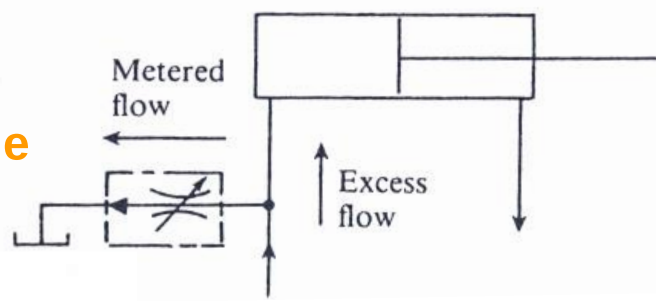
- Placement of the control. With the **METER OUT** configuration, there are less flow variations at the inlet and the fluid warming up before the cylinder is diminished.
- They are used to modify in a simple way the cylinder velocity when the pressure is more or less constant and there is no need for an accurate speed value (presses, working tables). Also, they avoid **pressure peaks (dampers)**.

PRESSURE line (meter in).



RETURN to tank line (meter out).

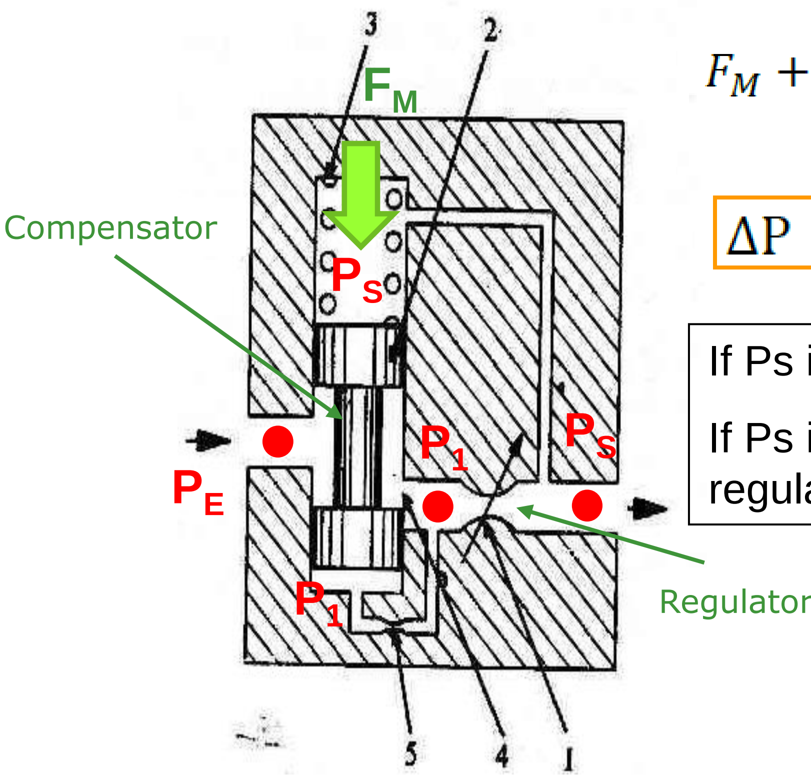




BLEED-OFF flow control.

Pressure compensated valves (I).

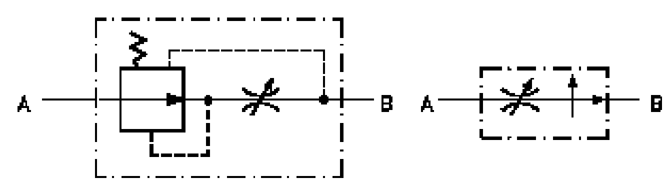
- They intend **constant pressure drop** at the regulator element.
- **A two orifice system**, one to adjust for load variations downstream and a second one for the flow control (relative position depends on actual valve).



$$F_M + P_S \cdot S = P_1 \cdot S \longrightarrow \boxed{\Delta P = P_1 - P_S = \frac{F_M}{S} = cte.}$$

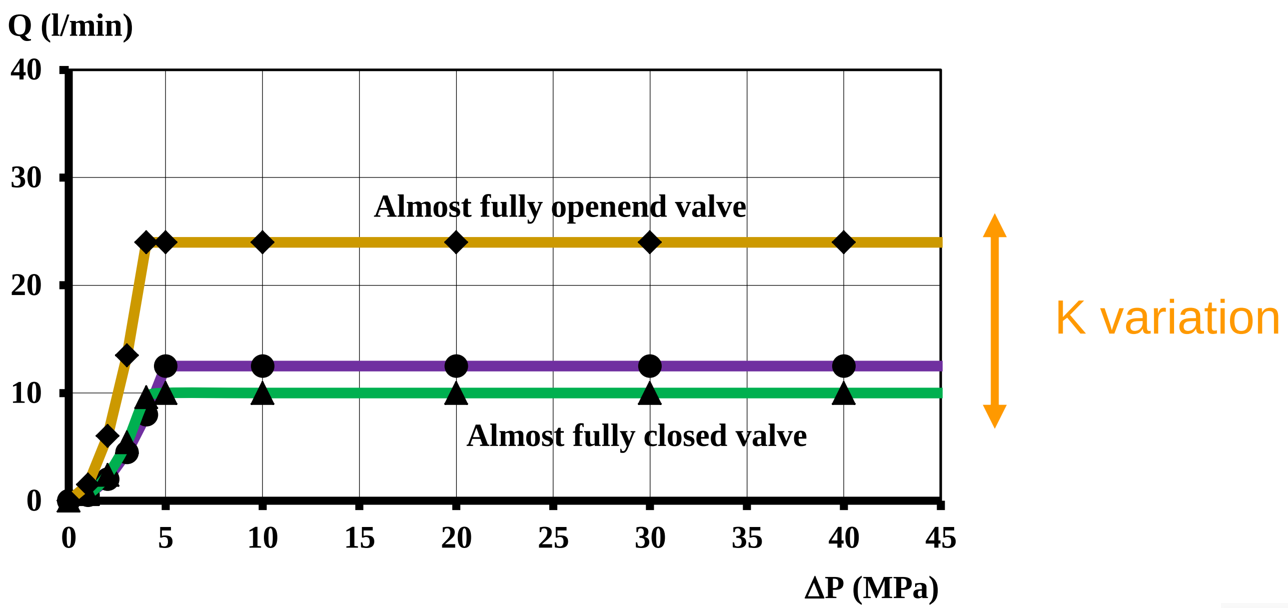
$$\boxed{\Delta P = K \cdot Q^2 = cte.}$$ **(Fixing K, Q is fixed)**

If Ps increases → ΔP decreases → Q will tend to a lower value
If Ps increases → Compensator opens→ Q increases (self-regulation)



Pressure compensated valves (III).

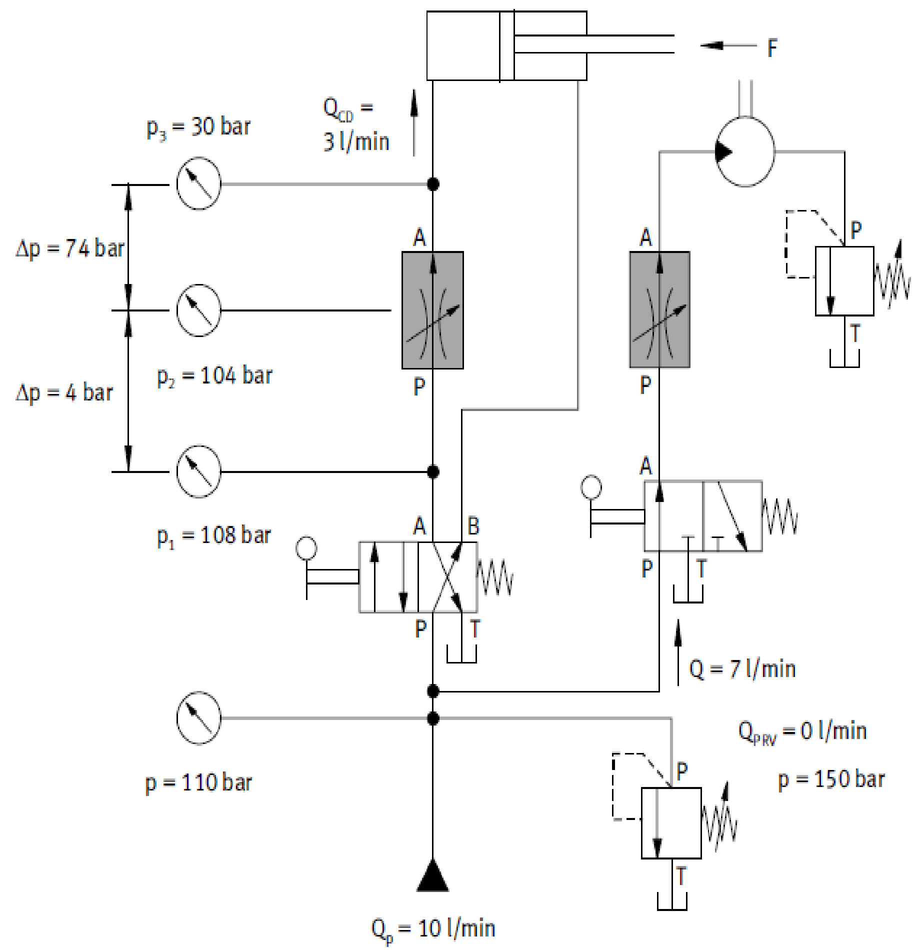
- Pressure drop in the restriction: 3 – 6 bar
- Minimum pressure drop in the valve: 5 – 12 bar
- Minimum flow rate to be controlled: 0.1 l/min
- Filtering needs: < 10 µm
- Compensation in temperature and/or viscosity (they have a metal stick that changes its length with temperature, modifying the area).



Exercise 6: UO: UOxxxHTU

At the system in the figure, the hydraulic motor moves a load with a torque of (Tx10+U) Nm and then, the fluid passes through a sequence valve at 5(H + 1) bar. Determine:

a) The minimum motor's cubic capacity, when the control of the two flow rate control valves (cylinder one compensated) in the extend is the same.
b) The force to be overcome at the extend if the total stroke is 10(5 + U) cm and it is done in (10 + T) s.
c) System's efficiency.
d) Braking time of the hydraulic motor if the inertial is 4 kg m².

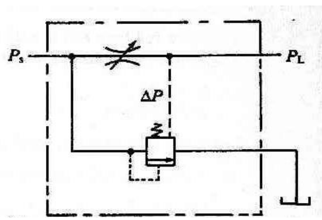


2.6. Flow control valves.

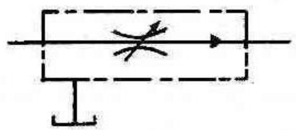
Pressure compensated flow control valves with tank connection (3 port valves).

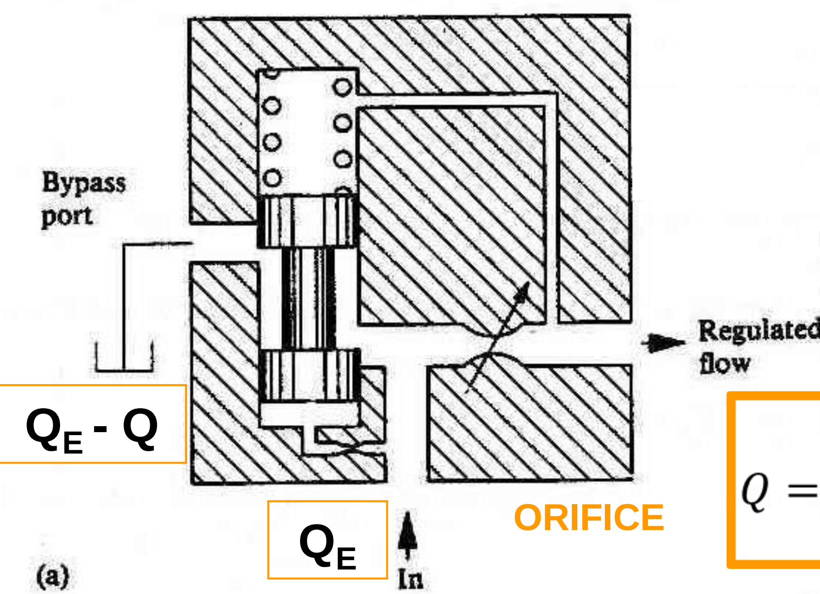
They include a relief valve to tank, making possible the real flow control and making it independent of the pressure in the system (load independent).

(detailed symbol)



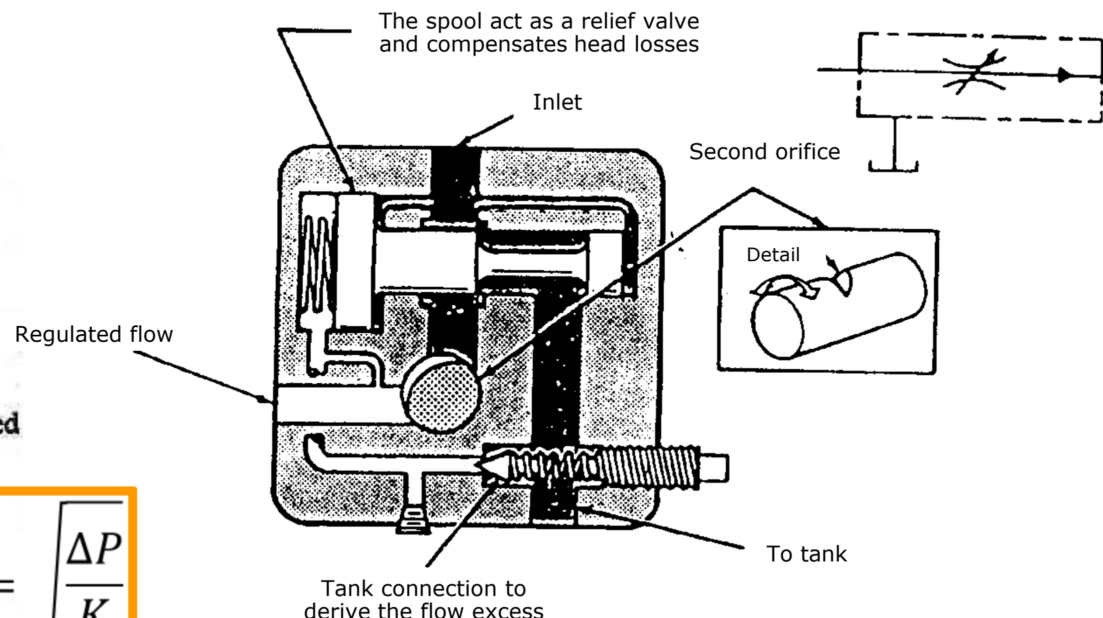
(simplified symbol)





ORIFICE

$$Q = \sqrt{\frac{\Delta P}{K}}$$

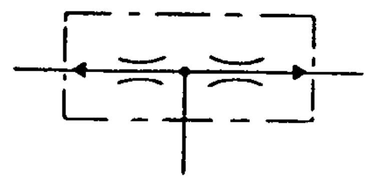


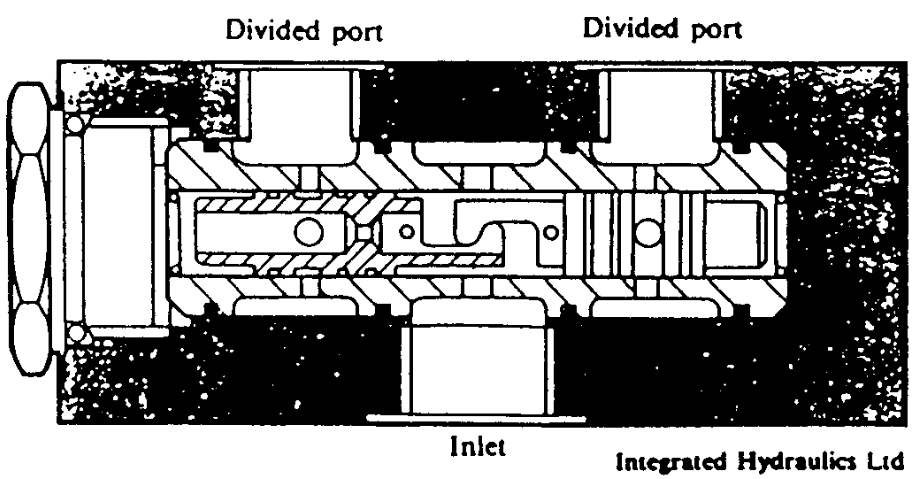
CYLINDER

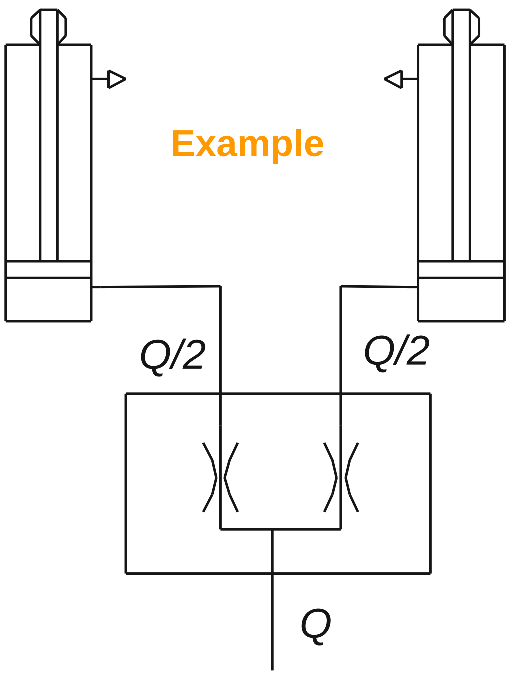
Flow dividers (I): valve type.

- Two **matched orifices** with interlocking spools, to split the flow (often) into to equal outputs. It is possible to obtain other ratios.
- Usage range: up to **200 l/min**

- **Sketch and symbol:**

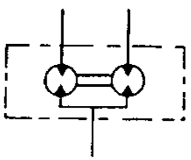






Flow dividers (II): Motor type.

- **Two or more motors** (usually gear motors) using the same shaft.
- If the motors have the **same cubic capacity**: equal division of the flow (except for possible volumetric losses).
- Act as **pressure intensifiers.**
- **Symbol and working principle:**

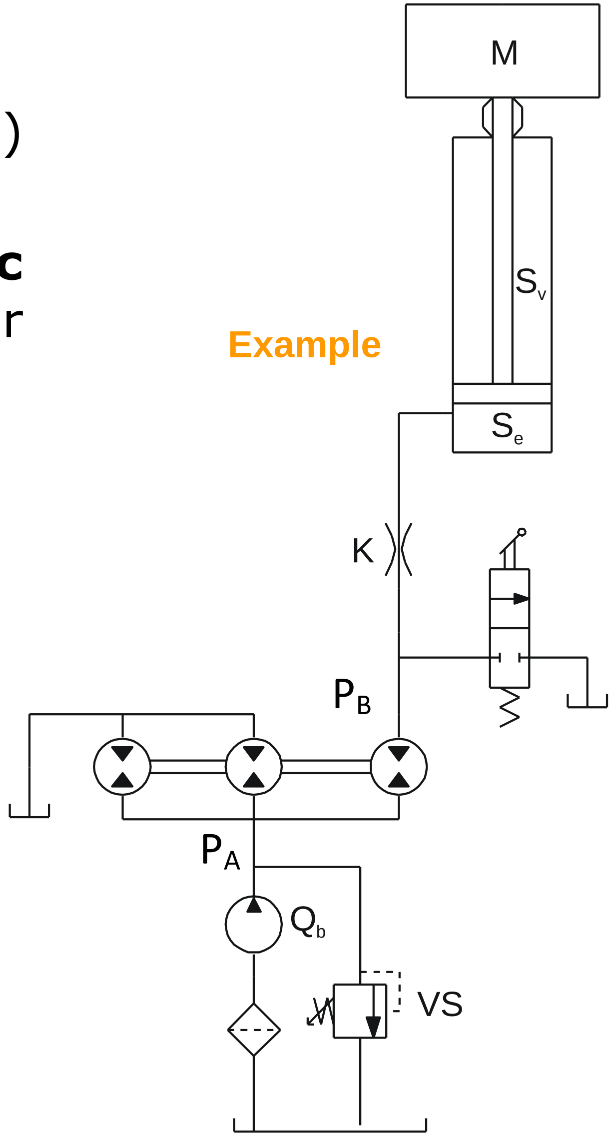


$$T_1 = T_2 = \frac{P_A \eta_H \vartheta_M}{2\pi}, \quad T_3 = \frac{(P_A - P_B)\eta_H \vartheta_M}{2\pi}$$

$$\sum T = 0 \quad \Rightarrow T_1 + T_2 + T_3 = 0 \Rightarrow \frac{(3P_A - P_B)\eta_H \vartheta_M}{2\pi} = 0$$

$$\boxed{P_B = 3P_A}$$

Example



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2.7. Other elements.

Filters (I).

- **Definition:** these are elements used in fluid power circuits to limit and isolate SOLID PARTICLES which are transported by the fluid (fluid pollution).

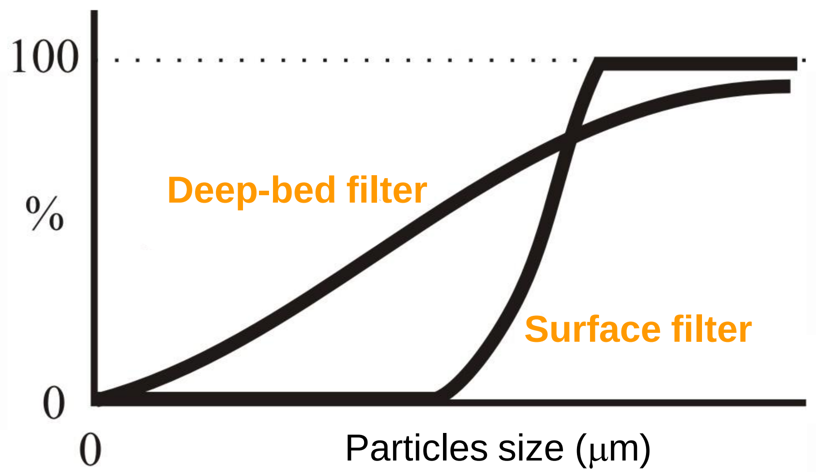
- **Needs for filtering:** The solid particles are harmful as their typical size is of the same order as the gaps and small pieces of the circuits, giving rise to EROSIONS and BLOCKAGE which may turn to bad working of the circuit and other secondary effects.
  - HIGH PRESSURE: Gaps around 5 microns.
  - LOW PRESSURE: Gaps between 16-40 microns.

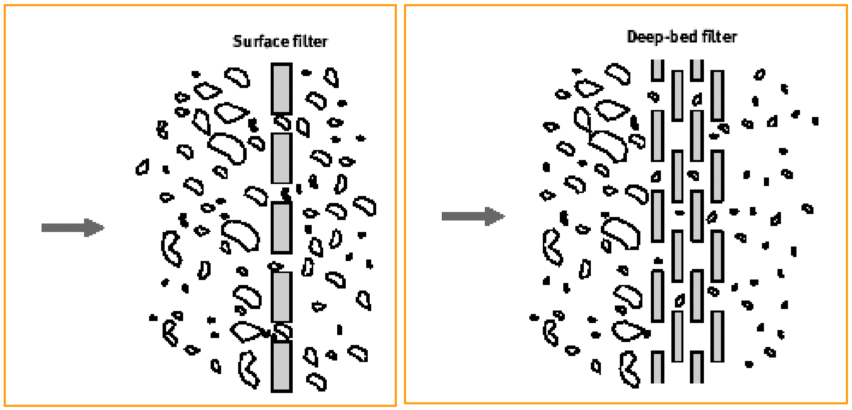
- **Effects of the fluid pollution:**
  - INTERFERE IN THE ENERGY TRANSFER: Blocking the relative movement of still and moving parts in radial gaps of pumps, cylinder walls, etc…
  - PIPE BLOCKAGE: Higher roughness (more losses). Less heat transmission (preventing cooling by boundary layer transmission).
  - HARMFUL FOR LUBRICATION: erosion is promoted, with material being detached and new pollution is generated
  - BLOCKAGE OF SMALL ELEMENTS: Valves, spools, flanges, etc…

Filters (II): Types.

- **SURFACE**:
  - It provides only one filtering layer. Often, made with a metallic layer or cellulose membrane or, even, porous ceramic materials.

- **DEEP-BED OR DEPTH**:
  - The fluid has to pass through a given portion of filtering material placed in a sandwich configuration between two metallic plates.

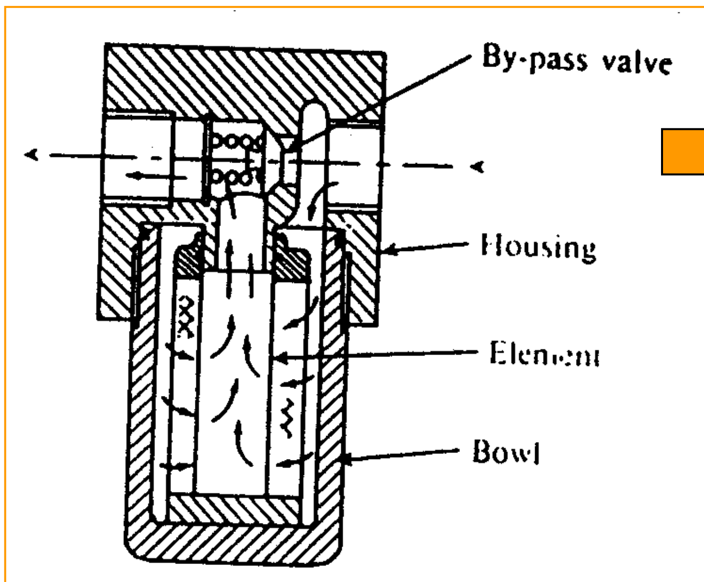




2.7. Other elements.

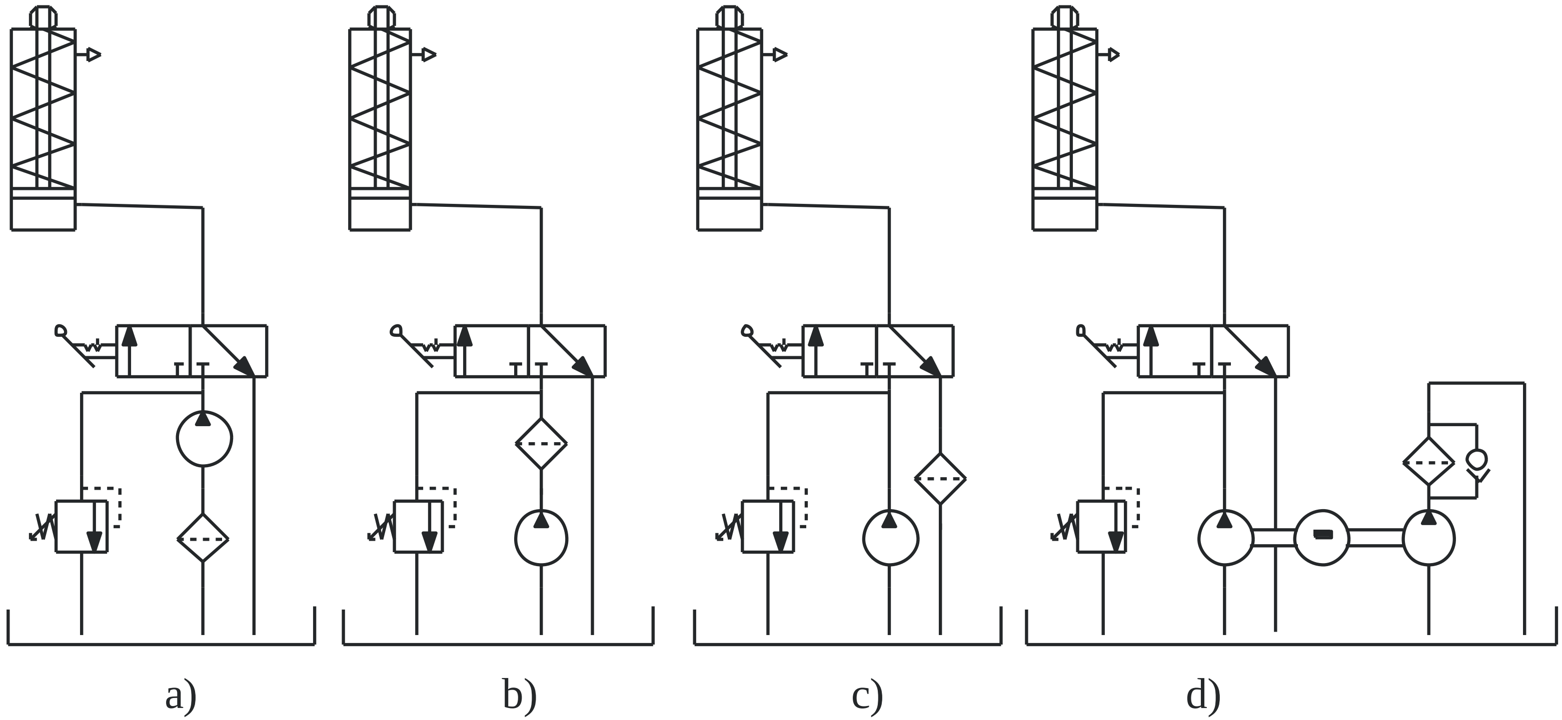
Filters (III): Design.

- **PRESSURE DROP:**
  - Maximum admissible: 10 bar.
  - Operative: 3-4 bar.
  - It increases with the particles being filtered.
  - Increasing with decreasing porosity.



If the filtering element becomes stuck or too dirty, the head loss becomes too big and the possible energetic implications or, even, cavitation problems may arise → A by-pass valve is often placed in parallel with the filter not to overpass an admissible value.

Filters (IV): Filter Locations and Efficiency.

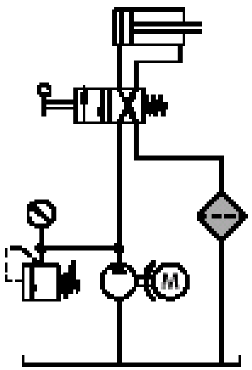
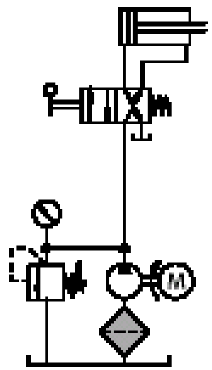
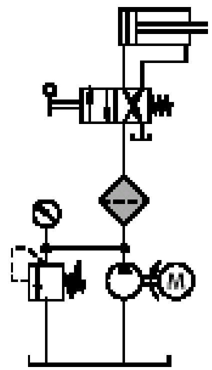
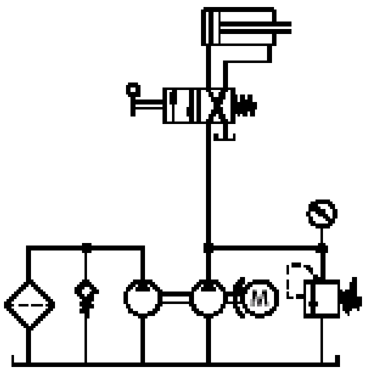


PUMP INLET | PRESSURE-LINE | RETURN TO TANK | CLEAN-UP LOOP FILTER

$$\beta_x = \frac{\text{number of particles upstream larger than x } \mu\text{m}}{\text{number of particles downstream larger than x } \mu\text{m}}$$

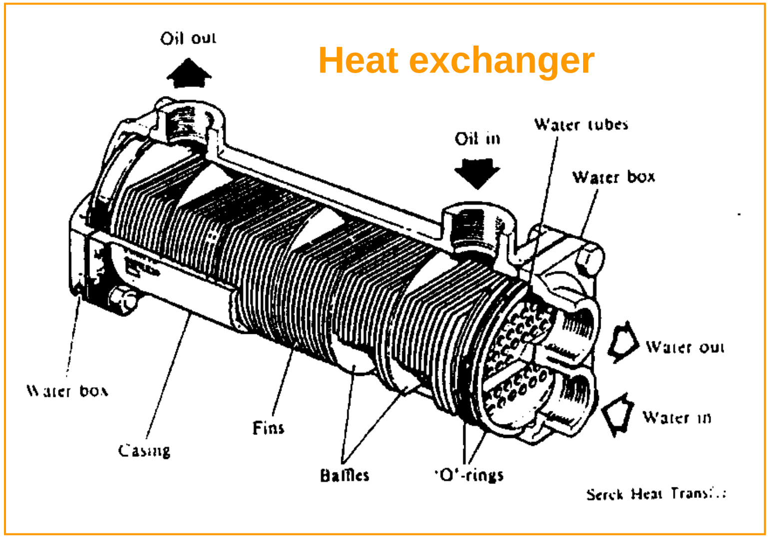
2.7. Other elements.

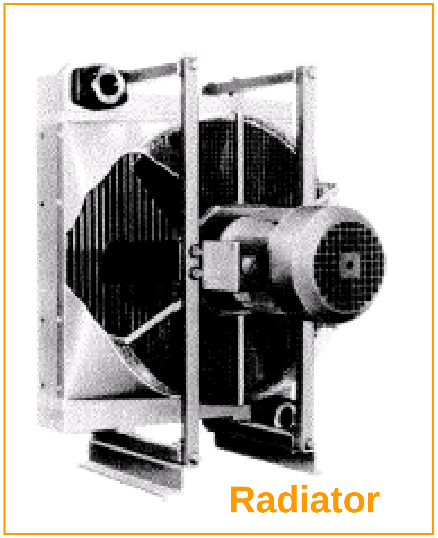
Filters (V): Situation.

	Filtering of the main flow			By-pass flow filtering
	Return flow filter	Pump inlet filter	Pressure line filter	
Circuit diagram				
Advantages	economical simple maintenance	protects pump from contamination	smaller pore size possible for valves sensitive to dirt	smaller filter possible as an additional filter
Disadvantages	contamination can only be checked having passed through the hydraulic components	difficult access, inlet problems with fine pored filters. Result: cavitation	expensive	lower dirt-filtering capacity
Remarks	frequently used	can also be used ahead of the pump as a coarse filter	requires a pressure-tight housing and contamination indicator	only part of the delivery is filtered

Heat exchangers and coolers.

- They are introduced in the system when the NORMAL COOLING by conduction in the tank becomes INSUFFICIENT (i.e. The foreseen heat exchange in the tank is not enough and the limit temperature for the fluid is reached).
- Very often used in systems with a clean-up loop filtering circuit (slide 51).
- Types:
  - **Heat exchange with a water-oil system or using plates or cross-flow tubes.**
  - **Radiator cooled by forced convention (fan).**

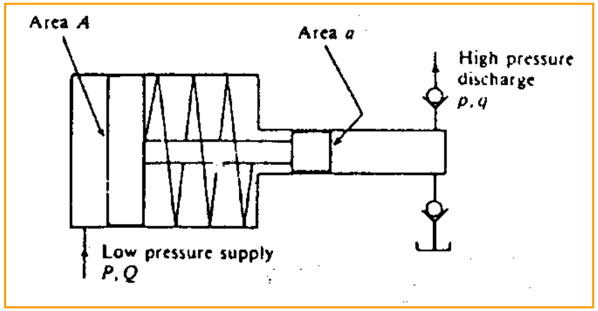




Pressure intensifiers.

- They are used to produce a high pressure at the outlet from a small pressure at the inlet. Inlet: **Pneumatic or oleo-hydraulic** and outlet: **Impulsive or continuous.**

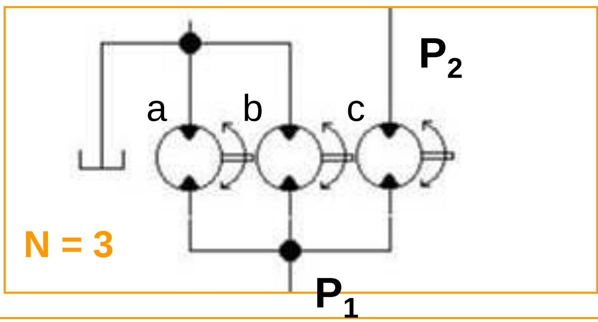
- **IMPULSE INTENSIFIERS:** Big size piston together with another with smaller diameter. The pressure increase is proportional to the surface quotient (the flow rate is reduced in an equivalent amount)



$$P = \frac{F}{A}, \quad p = \frac{F}{a}$$

$$p = P\frac{A}{a} \longrightarrow$$ **Pressure increase.**

- **CONTINUOUS INTENSIFIER:** Flow rate rotating division.



$$Q_a \Delta P_a + Q_b \Delta P_b + Q_c \Delta P_c = 0$$

$$Q\left(\Delta P_a + \Delta P_b + \Delta P_c\right) = 0 \Rightarrow \Delta P_c = -\Delta P_a - \Delta P_b$$

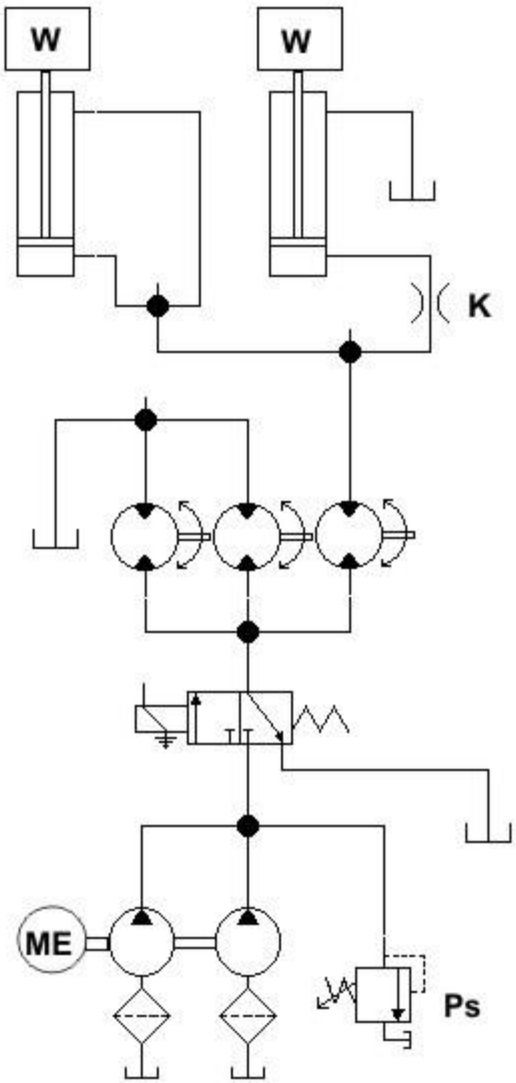
$$Como \quad \Delta P_a = \Delta P_b \quad y \quad -\Delta P_a = P_1$$

$$P_2 = P_1 + \Delta P_c = 3P_1 \longrightarrow$$ **Pressure increase.**

Exercise 7: UO: UOxxxHTU

The two cylinders of the figure are identical, with a circular piston/bore surface S_B = 100 cm^2, rod section S_R = 60 cm^2 and stroke L = 10(U + 1) cm and both stand the same load W = (200+ Hx10+T) kN. The valve constant (FCV) is k = (100 – 2H) Pa/(cm^3/s)^2. The pumps are twin ones, with ϑ = 8.33 cm^3/rev and are being moved at N = 1800 rpm. Determine:

a) Time evolution of the speeds in both cylinders and the pressure at the pumps' outlet from the starting up of the movement, by the action on the directional valve.
b) Same thing for the retract stroke (same load W).
c) Dissipated power in the flow rate control valve either in the advance and retract strokes.
d) What is the power required at the electrical mover?



Pipes and hoses

- Hydraulic oil is guided through the system through **pipes** (rigid joints) or **hoses** (moving or vibrating elements).
- The diameter should not be too small (noise, cavitation, pressure losses, heat generation, wear and aging ...) or too large (cost and space).
- The **diameter** (inside) is established from the **maximum flow** and the **average speed**:

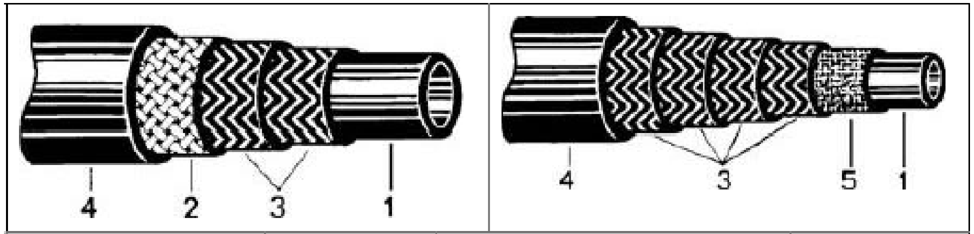
$$D = \sqrt{\frac{4 \cdot Q_{max}}{\pi \cdot v}}$$

- Recommended **average speeds** are as follows:

Pipes	Hoses
- Pressure lines **2 to 6 m/s**	- Pressure lines **2.1 to 4.6 m/s**
- Suction and return lines **0.6 to 1.6 m/s**	- Suction and return lines **0.6 to 1.2 m/s**

Pipes and hoses

- The pipes are usually mainly **carbon steel**.
- The hoses are made up of a **series of layers** that ensure adequate rigidity and reliability:



Inner Diameter		External Diameter	Pressure (bar)		Minimum Radius
mm	inch	mm	Operating	Rupture	mm
9.52	3/8	21.4	350	1400	125
12.7	1/2	24.6	280	1100	180
19.1	3/4	31.7	210	850	240
25.4	1	39.7	210	850	305

1. Synthetic rubber tube, resistant to oils and hydrocarbons. Typical temperature range: 40ºC-120ºC
2. Abrasion resistant metal screen
3. Mesh with braided wire and anti-friction rubber
4. Abrasion resistant synthetic rubber cover
5. Textile layer

- Once the diameter has been estimated using the previous formula, it will be necessary to refer to the technical catalogs of the manufacturers and see the **commercial diameters** available.